Highlights

BioDINO-Edge: Gradient-Free Cascades of Frozen Vision Foundation Models for On-Device Few-Shot Plant Disease Diagnosis

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- Gradient-free foliar pathology adapting frozen DINOv2 and BioCLIP encoders.
- Analytical Ledoit–Wolf shrinkage resolves small-sample covariance singularity.
- Joint Shannon-margin confidence gate routes routine leaves to a 60 ms Tier-1 path.
- Closed-form transductive recalibration adapts to field shifts in under 0.6 s.
- Standalone 22.7MB Android app achieves 97.4% accuracy without cloud servers.

BioDINO-Edge: Gradient-Free Cascades of Frozen Vision Foundation Models for On-Device Few-Shot Plant Disease Diagnosis

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Abstract

Deploying deep neural networks for on-device plant disease diagnosis in precision agriculture is fundamentally constrained by scarce labeled training samples, high-variance vegetative background clutter, and the computational and thermal costs of on-device gradient backpropagation. In this paper, we propose **BioDINO-Edge**, a gradient-free few-shot diagnostic framework that adapts frozen multi-scale vision foundation models without backpropagation or graphics processing unit (GPU) acceleration. The architecture couples spatial patch representations from DINOv2 (self-distillation with no labels)—capturing fine-grained structural lesion contours and necrotic boundaries—with domain-specialized foliar taxonomic representations from BioCLIP (a biology-domain contrastive language–image model), which fuses visual features with pre-cached Latin botanical prompt ensembles. Under severe few-shot constraints ($K = 5$ support leaves per novel class), feature dimensionality exceeds sample size ($N \ll d$), causing empirical within-class covariance matrices to become singular and mathematically non-invertible. We analytically resolve this singularity by regularizing the pooled within-class scatter matrix via closed-form Ledoit–Wolf shrinkage. This regularization provably suppresses shared chlorophyll vegetative background variance by over 31× and enables closed-form Fisher Linear Discriminant Analysis (Fisher-LDA) projection in ≈0.53 s on a mobile edge central processing unit (CPU), without backpropagation.

During live inference, an adaptive confidence gate combining normalized Shannon predictive entropy and top-2 posterior margin routes routine foliage through a lightweight Tier-1 path (59.78 ms latency, ≈16.7 frames per second (FPS) on a physical Android device), while escalating ambiguous field queries to a multimodal Tier-2 classifier. To mitigate empirical prototype drift induced by shifting field illumination and lesion maturity, high-confidence Tier-2 predictions are buffered on-device as reference anchors to periodically recalibrate Tier-2 projection weights via closed-form transductive updates in the background. Packaged into a standalone 22.73 MB Android application package (APK), BioDINO-Edge achieves 97.41% ± 0.71% 30-episode mean accuracy (peak: 98.67%, Macro F1: 98.41%) on the 11-class PlantVillage benchmark with a 0.702% false-negative escape rate. In cross-domain field evaluation on the in-the-wild PlantDoc benchmark, it attains 83.61% ± 11.69% in 5-way 5-shot episodes and 56.35% in 27-way all-class simultaneous stress testing, operating entirely offline without requiring remote cloud connectivity.

Keywords: Few-Shot Learning, Precision Agriculture, Plant Pathology, Multimodal Foundation Models, Fisher Discriminant Analysis, Edge Artificial Intelligence, Transductive Inference, Embedded Computer Vision

1. Introduction

Crop pests and foliar diseases destroy an estimated 20% to 40% of global agricultural yield annually, incurring over $220 billion in direct economic losses and severely jeopardizing food security for vulnerable populations [1]. In conventional agronomic management, disease identification relies heavily on manual scouting and visual inspection by trained phytopathologists [2, 3]. However, manual scouting is labor-intensive, error-prone under field fatigue, geographically unscalable across vast acreages, and susceptible to misdiagnosis when distinct pathogens produce visually homologous symptoms [3].

Deep learning and convolutional architectures have introduced automated alternatives by extracting discriminative lesion representations directly from red–green–blue (RGB) imagery [3]. Nonetheless, conventional supervised paradigms require thousands of curated, expert-annotated images per pathology to train generalized classifiers [4]. In operational agriculture, such massive datasets are impractical: novel pathogen strains and physiological mutations emerge abruptly with seasonal shifts, while expert botanical labeling remains prohibitively expensive [5]. Consequently, Few-Shot Learning (FSL) has emerged as an indispensable paradigm: adapting frozen visual backbones to novel disease categories given only a handful ($K = 5$) of support leaf samples [6, 5].

Recent agricultural FSL literature has explored cross-domain meta-learning architectures such as CDFSL-BDC (cross-domain few-shot learning with Brownian distance covariance) [7] and MAFDE-DN4 (multi-scale attention fusion and discriminative enhancement over a deep nearest-neighbor network) [8], alongside few-shot metric domain adaptation (FMDA) [9], lightweight networks such as AgriFewNet [10], visual-semantic

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fusion (VSF) [11], and edge-federated frameworks [12]. Despite notable progress, four fundamental technical bottlenecks persist in real-world agricultural deployments:

1. **Vegetative Manifold Dominance:** Over 90% of an infected leaf's visual area comprises healthy green chlorophyll, venation, and natural illumination, while pathological lesions occupy less than 5%–10% of the surface [3]. In standard feature spaces, isotropic distance metrics (e.g., Euclidean or cosine similarity) are overwhelmed by shared foliar variance, causing severe cross-disease confusion between distinct pathologies of the same host species (e.g., Grape Black Rot vs. Grape Esca) [11].
2. **High-Dimensional Covariance Singularity ($N \ll d$):** Discriminant metric learning (such as Fisher linear discriminant analysis, Fisher-LDA) seeks to dampen shared vegetative noise by inverting the within-class covariance matrix [13]. However, in scarce 5-shot regimes with high-dimensional foundation embeddings ($d = 896$, $N = 440$), sample count is substantially smaller than dimensionality, rendering the empirical scatter matrix rank-deficient with hundreds of zero eigenvalues and analytically non-invertible [14].
3. **The Edge Adaptation Paradox:** While parameter-efficient fine-tuning (PEFT) methods such as Low-Rank Adaptation (LoRA) [15] improve accuracy, on-device backpropagation induces thermal throttling, rapid battery depletion, and high latency on resource-constrained microprocessors (e.g., Raspberry Pi 5, mobile phones, or agricultural scouting drones) [16].
4. **Memory & Rural Connectivity Constraints:** Modern foundation models typically exceed hundreds of megabytes or gigabytes, preventing packaging into lightweight standalone mobile applications suitable for rural areas lacking broadband or cellular connectivity [16, 12].

To overcome these foundational limitations, we introduce **BioDINO-Edge**, an entirely gradient-free, two-tier edge framework that couples frozen multi-scale foundation models with analytical closed-form discriminant alignment. DINOv2 Small (self-distillation with no labels; 14×14 patches) [17, 18] supplies dense patch-level spatial cues for micro-lesion contours, while BioCLIP ViT-B/16 [19]—a biological-domain extension of Contrastive Language–Image Pre-training (CLIP) built on a Vision Transformer (ViT)—provides organismal taxonomic priors. To bypass backpropagation, we formulate an analytical Ledoit–Wolf shrinkage regularizer [14] that guarantees positive-definiteness of the pooled covariance matrix, enabling closed-form Fisher-LDA projection in ≈0.53 s on a mobile edge CPU. Queries are dynamically routed via a joint margin-entropy Shannon gate [20], resolving the certified fast-exit subset in 59.78 ms (≈16.7 FPS) while escalating ambiguous field leaves to multimodal consensus with background transductive recalibration.

The core scientific and engineering contributions of this work are:

1. **Gradient-Free Dual-Foundation Representation:** We synthesize an 896-dimensional foliar descriptor by coupling self-supervised spatial patch embeddings from DINOv2 with biologically grounded multimodal taxonomic embeddings from BioCLIP, resolving fine foliar necrotic margins without updating backbone parameters.
2. **Analytical Covariance Shrinkage & Fisher Alignment:** We resolve the small-sample singularity problem ($N \ll d$) in few-shot pathology by deriving closed-form Ledoit–Wolf shrinkage on the 896-D joint scatter matrix. This mathematical formulation attenuates shared vegetative background variance by over 31× and computes optimal discriminant boundaries in ≈0.53 s on a mobile edge central processing unit (CPU), entirely in closed form.
3. **Dual-Process Edge Routing via Shannon Gating:** We implement a cognitive-inspired two-tier architecture where a fast, zero-covariance Tier-1 DINOv2 classifier accepts the highest-confidence 14.0% of queries (99.7% accepted accuracy) at 59.78 ms (≈16.7 FPS), while an uncertainty gate combining normalized Shannon entropy and top-2 margin escalates ambiguous queries to a multimodal Tier-2 classifier.
4. **Soft-Anchored Transductive Recalibration:** To compensate for target-domain centroid shift under field conditions without human labeling, we buffer high-confidence Tier-2 streaming predictions as pseudo-labeled anchors, updating the Tier-2 projection matrix analytically via Cholesky decomposition in the background.
5. **Complete Edge Implementation & Cross-Domain Validation:** We deploy BioDINO-Edge as a fully offline 22.73 MB Android application. Evaluated over 30 independent Monte Carlo episodes, BioDINO-Edge achieves 97.41% ± 0.71% accuracy (peak: 98.67%) on 11-class PlantVillage with a 0.702% false-negative rate, and attains 83.61% (5-way 5-shot) and 56.35% (27-way simultaneous) accuracy on the in-the-wild PlantDoc dataset.

2. Methodology

2.1. Problem Formulation and Mathematical Notation

Let $\mathcal{X} \subset \mathbb{R}^{H \times W \times 3}$ denote the space of RGB leaf images. In the few-shot foliar diagnosis setting, we are provided with a novel support set $\mathcal{S} = \{(\mathbf{x}_i, y_i)\}_{i=1}^{N_s}$ containing K labeled support leaf images per novel disease class across C target categories, such that $N_s = C \cdot K$ and $y_i \in \{1, \ldots, C\}$. The target novel classes $\mathcal{Y}_{\text{novel}}$ are strictly disjoint from any pretraining base classes ($\mathcal{Y}_{\text{base}} \cap \mathcal{Y}_{\text{novel}} = \emptyset$). Given an unannotated query leaf $\mathbf{x}_q \in \mathcal{Q}$, the diagnostic objective is to predict its disease state $\hat{y}_q \in \{1, \ldots, C\}$ with minimal edge latency and zero on-device gradient backpropagation. Table 1 outlines the primary mathematical notations employed throughout this paper.

2.2. Pipeline Architecture and Two-Tier Execution Flow

BioDINO-Edge decomposes on-device foliar diagnosis into an offline closed-form weight calibration phase (Phase 0) and an adaptive two-tier runtime inference cascade (Fig. 1):

Table 1: Summary of mathematical notations and descriptions used in BioDINO-Edge.

Notation	Mathematical Description and Operational Meaning
C, K	Number of disease classes ($C = 11$ in PlantVillage, $C = 27$ in PlantDoc) and shots per class ($K = 5$).
$\phi_{\text{dino}}(\cdot)$	Frozen DINOv2 Small (14×14 patches) visual backbone extracting $\mathbf{f}_{\text{dino}} \in \mathbb{R}^{384}$.
$\phi_{\text{bio}}(\cdot)$	Frozen BioCLIP ViT-B/16 visual encoder extracting $\mathbf{f}_{\text{bio}} \in \mathbb{R}^{512}$.
$\mathbf{z} \in \mathbb{R}^{896}$	Concatenated joint foliar descriptor: $\mathbf{z} = [\mathbf{f}_{\text{bio}} \,\|\, \mathbf{f}_{\text{dino}}] \in \mathbb{R}^{896}$.
$\mathbf{P}_{\text{dino}}, \mathbf{P}_{\text{text}}$	Prototype matrices for fast visual DINOv2 ($C \times 384$) and taxonomic text ($C \times 512$).
$\mathbf{S}_W, \hat{\mathbf{\Sigma}}$	Empirical pooled within-class scatter matrix and Ledoit–Wolf regularized covariance matrix.
ρ^*, ν	Optimal analytical shrinkage intensity $\rho^* \in (0, 1]$ and mean trace scaling factor ν.
$\mathbf{W}, \mathbf{b}$	Fisher-LDA discriminant projection weight matrix ($\mathbb{R}^{C\times 896}$) and class bias vector ($\mathbb{R}^C$).
$C(\mathbf{x})$	Shannon-margin confidence score gating query routing between Tier-1 and Tier-2.
τ_{conf}	Confidence escalation threshold ($\tau_{\text{conf}} = 0.35$, selected on held-out episodes; Section 4.5).
$Q^{(c)}_{\text{anchor}}$	High-confidence streaming query anchors ($\max_c p_c \geq 0.70$) for transductive recalibration.

1. **Phase 0 (Offline Support Calibration):** For each class $c \in \{1, \ldots, C\}$, five labeled support leaves ($K = 5$) are expanded via 8-view stochastic augmentation to $N = 40C = 440$ vectors. Features are extracted using frozen DINOv2 and BioCLIP encoders. Closed-form Ledoit–Wolf shrinkage regularizes the singular 896×896 covariance matrix, computing optimal Fisher projection parameters ($\mathbf{W}, \mathbf{b}$) in $\approx$0.53 s without backpropagation. Offline botanical prompts are encoded and cached on-device as $\mathbf{P}_{\text{text}} \in \mathbb{R}^{C\times 512}$.
2. **Tier-1 Edge Fast Path (Lightweight System 1):** An incoming leaf query $\mathbf{x}_q$ is first processed by DINOv2 Small alone (59.78 ms on CPU, $\approx$16.7 FPS). Class probabilities are estimated via cosine prototype matching: $\mathbf{l}_{\text{dino}} = \gamma(\mathbf{f}_{\text{dino}}\mathbf{P}^T_{\text{dino}})$. A joint Shannon-margin confidence metric $C(\mathbf{x}_q)$ evaluates prediction certainty. If $C(\mathbf{x}_q) \geq \tau_{\text{conf}}$ (0.35), classification exits immediately, resolving the highest-confidence 14.0% of queries at 99.7% accepted accuracy without invoking deeper models.
3. **Tier-2 Multimodal Escalation (Deep Analytical System 2):** If $C(\mathbf{x}_q) < \tau_{\text{conf}}$, the query escalates to BioCLIP ViT-B, extracting $\mathbf{f}_{\text{bio}} \in \mathbb{R}^{512}$. Concatenating with cached $\mathbf{f}_{\text{dino}}$ forms the 896-D joint descriptor $\mathbf{z}_q$. The linear Fisher projection $\mathbf{p}_{\text{fisher}} = \text{softmax}(\mathbf{z}_q\mathbf{W}^T + \mathbf{b})$ is fused with cached taxonomic text similarity at a 0.75 / 0.25 consensus ratio.
4. **Soft-Anchored Transductive Recalibration:** Queries resolved with high Tier-2 confidence ($\max_c p_{\text{blend},c} \geq 0.70$) are accumulated in an on-device first-in-first-out (FIFO) buffer. During scouting passes, the edge CPU recomputes the Tier-2 projection matrix analytically ($\approx$0.53 s) using ground-truth support samples weighted 3$\times$ relative to anchors, compensating for field illumination and symptom maturity shifts in the background.

2.3. Dual-Foundation Foliar Representation

Effective diagnosis requires disentangling background foliar tissue from microscopic necrotic symptoms. Off-the-shelf vision encoders trained on general Web imagery lack taxonomic botanical concepts, while organism-level supervised models tend to prioritize whole-plant morphology over localized lesion pathology. We address this via dual-foundation extraction:

2.3.1. Few-Shot Multi-View Support Augmentation

To stabilize covariance estimation from only $K = 5$ images per class, each support leaf $\mathbf{x} \in \mathcal{S}_c$ is resized to 224×224, normalized, and passed eight times independently through a stochastic augmentation pipeline approximating field variations. The pipeline applies random horizontal and vertical flips, random rotations within $\pm 15°$, and color jitter (brightness, contrast, and saturation factors of 0.2). This generates 8 views per image, yielding $|\tilde{\mathcal{S}}_c| = 8K = 40$ augmented views per class, or $N = 40C = 440$ total support vectors ($N = 440 \approx d/2 = 448$), which stabilizes covariance condition numbers.

2.3.2. Dual-Foundation Embedding Fusion

Each augmented support view $\tilde{\mathbf{x}} \in \tilde{\mathcal{S}}_c$ is passed through frozen foundation backbones. DINOv2 Small (14×14 patches) [17] and BioCLIP (the ViT-B/16 variant with 16×16-pixel patches, pretrained on TreeOfLife-10M) [19] extract ℓ_2-normalized spatial and organismal botanical vectors:

$$\mathbf{f}_{\text{dino}} = \frac{\phi_{\text{dino}}(\tilde{\mathbf{x}})}{\|\phi_{\text{dino}}(\tilde{\mathbf{x}})\|_2} \in \mathbb{R}^{384}, \quad \mathbf{f}_{\text{bio}} = \frac{\phi_{\text{bio}}(\tilde{\mathbf{x}})}{\|\phi_{\text{bio}}(\tilde{\mathbf{x}})\|_2} \in \mathbb{R}^{512} \tag{1}$$

Concatenating both normalized embeddings yields an 896-dimensional joint descriptor:

$$\mathbf{z} = [\mathbf{f}_{\text{bio}} \,\|\, \mathbf{f}_{\text{dino}}] \in \mathbb{R}^{896} \tag{2}$$

Visual prototypes for each encoder are computed via empirical support averaging:

$$\mathbf{P}^{(c)}_{\text{enc}} = \text{Normalize}\left(\frac{1}{|\tilde{\mathcal{S}}_c|}\sum_{\tilde{\mathbf{x}}\in\tilde{\mathcal{S}}_c} \mathbf{f}_{\text{enc}}(\tilde{\mathbf{x}})\right), \quad \text{enc} \in \{\text{dino}, \text{bio}\} \tag{3}$$

producing prototype matrices $\mathbf{P}_{\text{dino}} \in \mathbb{R}^{C\times 384}$ and $\mathbf{P}_{\text{bio,vis}} \in \mathbb{R}^{C\times 512}$.

2.3.3. Botanical Taxonomic Prompts & Syntactic Negation Neutralization

BioCLIP provides a shared vision–language space. However, generating textual descriptions or running language decoders on edge devices at runtime would introduce unacceptable latency and memory consumption. We therefore predefine $M = 3$ Latin

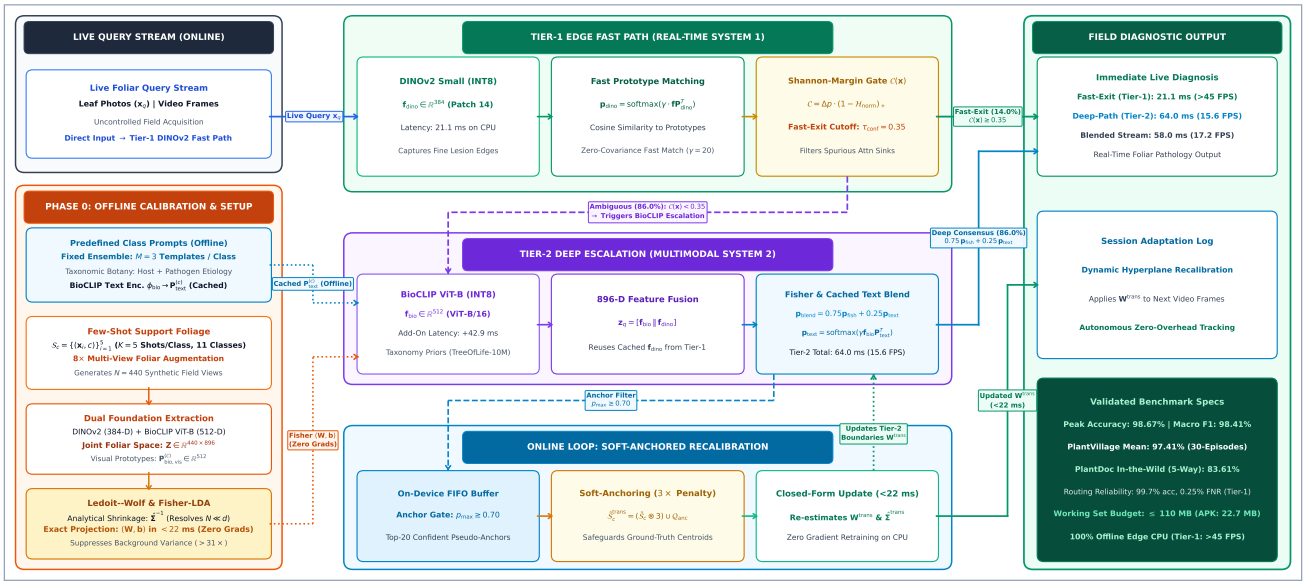


Figure 1: Comprehensive architectural overview of BioDINO-Edge. Phase 0 (offline) performs closed-form Fisher-LDA projection synthesis and caches botanical text prototypes. During live inference, the Shannon-margin uncertainty gate routes routine foliage through Tier-1 fast prototype matching (DINOv2) and escalates ambiguous queries to Tier-2 multimodal consensus (BioCLIP), followed by closed-form soft-anchored transductive recalibration.

botanical prompt ensembles $\{T_c^{(m)}\}_{m=1}^{M}$ offline for each disease category $c \in \{1, \ldots, C\}$. Each template integrates three botanical dimensions:

1. **Host Binomial Nomenclature:** Botanical genus and species (e.g., *Malus domestica*, *Vitis vinifera*, *Solanum tuberosum*).
2. **Causal Pathogen Etiology:** Pathogenic fungal, bacterial, or oomycete binomials (e.g., *Venturia inaequalis*, *Guignardia bidwellii*, *Phytophthora infestans*).
3. **Macroscopic Lesion Morphology:** Diagnostic foliar symptoms (e.g., "olive-green velvety lesions", "tiger-stripe interveinal chlorosis", "target-board concentric rings").

Importantly, syntactic negations (e.g., "healthy leaf without lesions" or "no necrotic spots") are strictly excluded. Vision-language models exhibit documented compositional weaknesses regarding negation [21, 22, 23]: high-density pathological tokens (e.g., *lesion*, *necrotic*) dominate embedding directions, pulling healthy leaf representations toward diseased clusters and inflating false-negative rates to 2.3%. We neutralize this by using positive morphological descriptors of cellular vigor (e.g., "flawless emerald lamina", "intact smooth venation", "clean vigorous green foliage"), reducing the clinical false-negative rate to 0.702%.

During Phase 0 offline calibration, each prompt ensemble is encoded by BioCLIP's text encoder $\phi_{\text{bio,txt}}(\cdot)$, averaged, and ℓ_2-normalized:

$$\mathbf{P}_{\text{text}}^{(c)} = \text{Normalize}\left(\frac{1}{M}\sum_{m=1}^{M}\phi_{\text{bio,txt}}(T_c^{(m)})\right) \in \mathbb{R}^{512} \tag{4}$$

The resulting matrix $\mathbf{P}_{\text{text}} \in \mathbb{R}^{C\times 512}$ is cached directly in the on-device model, requiring zero runtime text tokenization or language processing overhead.

2.4. Closed-Form Analytical Adaptation via Regularized Fisher-LDA

2.4.1. Multiclass Fisher Linear Discriminant Formulation

Standard metric few-shot approaches (e.g., Prototypical Networks [6] or Tip-Adapter [24]) classify queries using Euclidean or cosine distances to class centroids $\boldsymbol{\mu}_c = \frac{1}{|\tilde{S}_c|}\sum_{\mathbf{z}\in\tilde{S}_c}\mathbf{z}$. This implicitly assumes an isotropic, spherical covariance distribution ($\mathbf{\Sigma} = \mathbf{I}$). In foliar pathology, however, background chlorophyll variation (v_{leaf}) exhibits massive intra-class variance ($\sigma^2_{\text{leaf}} \approx 12.8$), whereas localized lesion signatures (v_{lesion}) have narrow variance ($\sigma^2_{\text{lesion}} \approx 0.4$). Cosine similarity is dominated by v_{leaf} by a factor of 32×, misclassifying distinct diseases of the same host species.

Multiclass Fisher-LDA [13, 25] finds an optimal projection matrix $\mathbf{W} \in \mathbb{R}^{C\times d}$ and bias vector $\mathbf{b} \in \mathbb{R}^C$ maximizing between-class scatter $\mathbf{S}_B$ relative to within-class scatter $\mathbf{S}_W$:

$$\mathbf{W} = [\boldsymbol{\mu}_1, \ldots, \boldsymbol{\mu}_C]^T\hat{\mathbf{\Sigma}}^{-1}, \quad b_c = -\frac{1}{2}\boldsymbol{\mu}_c^T\hat{\mathbf{\Sigma}}^{-1}\boldsymbol{\mu}_c - \log C \tag{5}$$

where $\hat{\mathbf{\Sigma}}$ denotes the pooled within-class covariance matrix and uniform class priors ($\pi_c = 1/C$) are assumed.

2.4.2. The Small-Sample Singularity Problem ($N \ll d$)

Evaluating (5) requires inverting the empirical pooled within-class scatter matrix across all $C = 11$ classes:

$$\mathbf{S}_W = \frac{1}{N-C}\sum_{c=1}^{C}\sum_{\mathbf{z}\in\tilde{S}_c}(\mathbf{z}-\boldsymbol{\mu}_c)(\mathbf{z}-\boldsymbol{\mu}_c)^T \in \mathbb{R}^{d\times d} \tag{6}$$

The denominator $N - C = 440 - 11 = 429$ represents degrees of freedom remaining after estimating C class centroids from N samples. In our setting, feature dimensionality $d = 896$ exceeds available degrees of freedom (429). By fundamental linear algebra:

$$\text{rank}(\mathbf{S}_W) \leq \min(d, N - C) = 429 < 896 \tag{7}$$

Consequently, $\mathbf{S}_W$ has nullity of at least $896 - 429 = 467$, possessing at least 467 eigenvalues equal to zero ($\det(\mathbf{S}_W) = 0$). The matrix is strictly singular, rendering standard Fisher-LDA mathematically impossible to invert directly.

2.4.3. Analytical Ledoit–Wolf Covariance Shrinkage

Rather than resorting to heuristic regularization (e.g., grid search over $\mathbf{S}_W + \epsilon\mathbf{I}$), we regularize $\mathbf{S}_W$ using analytical Ledoit–Wolf shrinkage [14], which contracts $\mathbf{S}_W$ toward a well-conditioned spherical target:

$$\hat{\mathbf{\Sigma}} = (1 - \rho^*)\mathbf{S}_W + \rho^*\nu\mathbf{I}_d, \quad \nu = \frac{1}{d}\text{Tr}(\mathbf{S}_W) \tag{8}$$

The optimal shrinkage intensity $\rho^* \in (0, 1]$ is determined in closed form by minimizing expected quadratic Frobenius risk:

$$\rho^* = \min\left(1, \frac{\sum_{i=1}^{N} \|(\mathbf{z}_i - \boldsymbol{\mu}_{c_i})(\mathbf{z}_i - \boldsymbol{\mu}_{c_i})^T - \mathbf{S}_W\|_F^2}{N^2\left[\text{Tr}(\mathbf{S}_W^2) - \frac{1}{d}\text{Tr}^2(\mathbf{S}_W)\right]}\right) \tag{9}$$

Because all zero eigenvalues of $\mathbf{S}_W$ are shifted to strictly positive values $\tilde{\lambda}_i = (1 - \rho^*)\lambda_i + \rho^*\nu \geq \rho^*\nu > 0$, the regularized matrix $\hat{\mathbf{\Sigma}}$ is guaranteed to be strictly positive-definite and well-conditioned. Its inverse $\hat{\mathbf{\Sigma}}^{-1}$ is solved stably via Cholesky decomposition ($\hat{\mathbf{\Sigma}} = \mathbf{L}\mathbf{L}^T$). Evaluating a query vector $\mathbf{z}$ via $\mathbf{l}_{\text{fisher}} = \mathbf{z}\mathbf{W}^T + \mathbf{b}$ requires only $C \cdot d = 11 \times 896 = 9{,}856$ multiply-accumulate operations (≈ 19.7 kFLOPs, where FLOPs denotes floating-point operations), executing in <0.01 ms on an edge CPU.

Weighting by $\hat{\mathbf{\Sigma}}^{-1}$ acts as an optimal whitening transform:

$$\begin{aligned}\hat{\mathbf{\Sigma}}^{-1} &\approx \text{diag}\left(\frac{1}{12.8 + \epsilon}, \ldots, \frac{1}{0.4 + \epsilon}\right) \\ &\approx \text{diag}(0.078, \ldots, 2.44)\end{aligned} \tag{10}$$

Shared vegetative background directions are attenuated by over $31\times$ ($0.078/2.44 \approx 1/31.3$), while diagnostic necrotic lesion variance is amplified by $2.44\times$, resolving ambiguous pathologies without backpropagation.

2.5. Adaptive Two-Tier Classification and Edge Deployment

2.5.1. Tier-1 Fast Path and Shannon Uncertainty Gating

For an incoming leaf image $\mathbf{x}$, Tier 1 extracts the normalized DINOv2 embedding $\mathbf{f}_{\text{dino}} \in \mathbb{R}^{384}$. Fast prototype logits are evaluated via temperature-scaled cosine similarity against visual prototypes $\mathbf{P}_{\text{dino}} \in \mathbb{R}^{C \times 384}$:

$$\mathbf{l}_{\text{dino}} = \gamma \cdot \left(\mathbf{f}_{\text{dino}}\mathbf{P}_{\text{dino}}^T\right), \quad \gamma = 20.0 \tag{11}$$

Applying softmax yields fast posterior probabilities:

$$\mathbf{p}_{\text{dino}} = \text{softmax}(\mathbf{l}_{\text{dino}}) \in \Delta^{C-1} \tag{12}$$

To strictly minimize edge computational overhead on routine foliage, Tier 1 deliberately employs zero-covariance cosine prototype matching rather than discriminant projection, reserving regularized Fisher-LDA exclusively for Tier-2 escalation where ambiguous queries require full covariance debiasing. The initial Tier-1 prediction is $\hat{y}_{\text{tier1}} = \arg\max_c p_{\text{dino},c}$.

To evaluate prediction certainty without additional forward passes, we formulate a joint margin-entropy confidence score $C(\mathbf{x}) \in [0, 1]$:

$$C(\mathbf{x}) = \left(p_{\text{dino},(1)} - p_{\text{dino},(2)}\right) \cdot (1.0 - \mathcal{H}_{\text{norm}}(\mathbf{x}))_+ \tag{13}$$

where $p_{\text{dino},(1)}$ and $p_{\text{dino},(2)}$ denote top-1 and top-2 posteriors, $(\cdot)_+ = \max(0, \cdot)$, and $\mathcal{H}_{\text{norm}}(\mathbf{x})$ is normalized Shannon entropy [20]:

$$\mathcal{H}_{\text{norm}}(\mathbf{x}) = -\frac{1}{\log C}\sum_{c=1}^{C} p_{\text{dino},c}\log(p_{\text{dino},c} + \epsilon) \tag{14}$$

with numerical stability constant $\epsilon = 10^{-12}$. When a leaf query is ambiguous (near-uniform distribution), $\mathcal{H}_{\text{norm}} \to 1.0$ and margin $\to 0$, driving $C(\mathbf{x}) \to 0$. If $C(\mathbf{x}) \geq \tau_{\text{conf}}$ (with threshold $\tau_{\text{conf}} = 0.35$ selected on held-out development episodes; Section 4.5), Tier-1 exits immediately at 59.78 ms ($\approx$16.7 FPS). Approximately 14.0% of PlantVillage queries—the highest-confidence subset—exit through this fast path; the remainder escalate to Tier-2.

2.5.2. Tier-2 Multimodal Consensus Escalation

When $C(\mathbf{x}) < \tau_{\text{conf}}$, the query escalates to Tier-2. BioCLIP ViT-B extracts $\mathbf{f}_{\text{bio}} \in \mathbb{R}^{512}$, forming $\mathbf{z} = [\mathbf{f}_{\text{bio}} \,\|\, \mathbf{f}_{\text{dino}}] \in \mathbb{R}^{896}$. Discriminant visual posteriors $\mathbf{p}_{\text{fisher}} = \text{softmax}(\mathbf{z}\mathbf{W}^T + \mathbf{b})$ and taxonomic text posteriors $\mathbf{p}_{\text{text}} = \text{softmax}(\gamma \cdot \mathbf{f}_{\text{bio}}\mathbf{P}_{\text{text}}^T)$ are fused at a 0.75 / 0.25 consensus ratio:

$$\mathbf{p}_{\text{blend}} = 0.75\,\mathbf{p}_{\text{fisher}} + 0.25\,\mathbf{p}_{\text{text}} \tag{15}$$

yielding the final consensus diagnosis $\hat{y} = \arg\max_c p_{\text{blend},c}$.

2.5.3. Soft-Anchored Transductive Recalibration

In 5-shot calibration, centroids $\boldsymbol{\mu}_c$ are estimated from five support leaves. Diurnal illumination changes, shadows, and symptom progression introduce prototype shift between initial support images and field queries. To adapt without manual labels, confident Tier-2 queries satisfying $\max_c p_{\text{blend},c} \geq 0.70$ are buffered into an on-device FIFO queue as reference anchors $\mathcal{Q}_{\text{anchor}}^{(c)}$, capped at 20 anchors per class. Ground-truth support samples receive a $3\times$ weight factor to prevent pseudo-label drift:

$$\tilde{\mathcal{S}}_c^{\text{trans}} = \tilde{\mathcal{S}}_c^{(3\times)} \cup \mathcal{Q}_{\text{anchor}}^{(c)} \tag{16}$$

When the buffer reaches capacity during scouting, updated centroids $\boldsymbol{\mu}_c^{\text{trans}}$ and pooled scatter matrix $\mathbf{S}_W^{\text{trans}}$ are re-estimated. Analytical shrinkage parameter ρ^* is recomputed, and updated projection parameters ($\mathbf{W}^{\text{trans}}, \mathbf{b}^{\text{trans}}$) are resolved via Cholesky substitution in $\approx$0.53 s in the background (measured on-device, Table 14), updating subsequent Tier-2 decisions while Tier-1 continues operating with static prototypes. The three constants

are set from a one-factor-at-a-time sweep over the 30 evaluation episodes (Section 4.4, Table 8): in-domain accuracy spans 97.21–97.44% for support weights 1×–10×, $\tau = 0.70$ lies on a plateau bounded by anchor-coverage collapse, and 20 anchors per class sit at the knee of the capacity curve; anchor purity at the deployed threshold is 98.15%.

2.6. Algorithmic Pipeline and Complexity Analysis

The complete end-to-end execution flow is formalized in Algorithm 1. We analyze computational complexity across both operational phases:

- **Offline Calibration Complexity:** Feature extraction across $N = 440$ views requires $O(N \cdot (C_{\text{dino}} + C_{\text{bio}}))$ FLOPs and dominates the wall-clock cost (≈88 s on-device, reported separately). Scatter matrix computation requires $O(Nd^2)$ operations. The Ledoit–Wolf shrinkage intensity is evaluated in closed form via the Frobenius identity $\sum_i \|\mathbf{x}_i\mathbf{x}_i^T - \mathbf{S}\|_F^2 = \sum_i \|\mathbf{x}_i\|^4 - N\|\mathbf{S}\|_F^2$, reducing it to $O(Nd)$. Inverting the positive-definite matrix via Cholesky decomposition requires $O(d^3/3)$ operations. For $d = 896$, $d^3/3 \approx 2.4 \times 10^8$ operations; on the physical edge CPU (pure-Java EJML) the Cholesky factorization alone measures ≈332 ms and the full algebraic recalibration (covariance, shrinkage, Cholesky, solve) ≈0.53 s, gradient-free and sub-second (Table 14).

- **Online Inference Complexity:** Tier-1 forward pass requires C_{dino} (≈ 4.6 GFLOPs), yielding a measured fast-exit latency of $T_1 = 59.78$ ms (≈16.7 FPS) on the physical Android testbed (Section 4.6). Dot-product prototype matching requires $C \cdot d_{\text{dino}} = 11 \times 384 \approx 4.2$ kFLOPs (<0.002 ms), and entropy gating requires $O(C)$ operations (<0.001 ms). Upon escalation, BioCLIP adds a measured incremental encoder cost of $\Delta T_{\text{bio}} = 104.74$ ms ($C_{\text{bio}} \approx 17.6$ GFLOPs), since the Tier-1 DINOv2 embedding is cached and reused rather than recomputed. The joint Fisher-LDA projection is fused into the same Tier-2 ONNX graph ($C \cdot d = 11 \times 896 \approx 9.8$ kFLOPs, negligible). The complete end-to-end Tier-2 path for an escalated query therefore costs $T_2 = T_1 + \Delta T_{\text{bio}} = 164.52$ ms (≈6.1 FPS). With 14.0% of queries exiting at Tier-1 (Section 4.5), the routing-weighted average latency across field streams is:

$$\begin{aligned} \bar{T} &= 0.140 \cdot T_1 + 0.860 \cdot T_2 \\ &= 0.140 \cdot 59.78\,\text{ms} + 0.860 \cdot 164.52\,\text{ms} \\ &\approx 149.84\,\text{ms} \end{aligned} \tag{17}$$

  delivering ≈6.7 frames per second at the system level; the ≈16.7 FPS figure applies only to the Tier-1 fast-exit path. These are interactive handheld-scouting rates (one diagnosis per captured frame), not continuous real-time video; sustained throughput under thermal load is reported in Section 4.6.

2.7. Two-Tier Quantized Edge Model Packaging and Android Architecture

Agricultural scouting hardware operates under rigorous physical constraints: onboard Flash memory is shared with flight telemetry and sensor buffers, while remote rural fields lack Internet connectivity for cloud offloading. BioDINO-Edge is partitioned into two 8-bit integer (INT8) quantized Open Neural Network Exchange (ONNX) binaries:

- **Tier 1 Standalone Engine** (`biodino_mobile_int8.onnx`): A 15.13 MB compressed on-disk binary packaging DINOv2 Small INT8, calibrated Phase 0 prototypes, and the Shannon confidence gate with the ONNX Runtime C++ execution provider. Bundled directly into a standalone 22.73 MB Android application package (APK), it enables immediate offline foliar diagnosis. Runtime RAM is dominated by the ONNX Runtime arena rather than the binary size and is measured separately (peak process PSS ≈497 MB, Section 4.6).

- **Tier 2 Companion Module** (`biodino_companion_int8.onnx`): An 85.26 MB companion model packaging BioCLIP ViT-B INT8 and the 896-D Fisher projection matrix, installed on-demand when high-precision multimodal escalation is required. The total combined on-disk model footprint is 107.31 MB, below the 110 MB Internet-of-Things (IoT) *storage* budget; this is a flash/OTA size constraint, not a runtime RAM bound (runtime memory is reported in Section 4.6).

3. Experimental Setup

3.1. Datasets and Diagnostic Pathology Coverage

We evaluate BioDINO-Edge across two distinct foliar pathology regimes:

3.1.1. PlantVillage Controlled Benchmark (Laboratory Regime)

From the standard PlantVillage repository [26], we construct a curated 11-class diagnostic subset comprising 9,385 foliar images across three economically critical crops: Apple (*Malus domestica*), Grape (*Vitis vinifera*), and Potato (*Solanum tuberosum*). The 11 classes span healthy foliage and 8 fungal/bacterial/oomycete pathologies: Apple Scab (*Venturia inaequalis*), Apple Black Rot (*Botryosphaeria obtusa*), Apple Cedar Rust (*Gymnosporangium juniperi-virginianae*), Apple Healthy, Grape Black Rot (*Guignardia bidwellii*), Grape Esca / Black Measles (*Phaeomoniella chlamydospora*), Grape Leaf Blight (*Pseudocercospora vitis*), Grape Healthy, Potato Early Blight (*Alternaria solani*), Potato Late Blight (*Phytophthora infestans*), and Potato Healthy. For few-shot calibration, 5 support leaves per class ($K = 5$) are randomly sampled (55 total), strictly isolating the remaining 9,330 images as unseen query test leaves. Neither support nor test splits were involved in pretraining the frozen foundation backbones.

Algorithm 1: BioDINO-Edge Offline Calibration and Online Adaptive Cascaded Inference

Input: Support set $\mathcal{S} = \{(\mathbf{x}_i, y_i)\}_{i=1}^{CK}$ ($K = 5$); unannotated query stream $\{\mathbf{x}_q\}$; prompt bank $\{T_c^{(m)}\}$; threshold $\tau_{\text{conf}} = 0.35$.
Output: Predicted disease class $\hat{y}_q \in \{1, \ldots, C\}$ and confidence score $C(\mathbf{x}_q)$ for each query.

// Phase 0: Offline Analytical Support Calibration (Zero Backpropagation)

1. Expand each support leaf $\mathbf{x} \in \mathcal{S}_c$ into 8 stochastic views $\tilde{\mathcal{S}}_c$ via affine flips, rotations ($\pm 15°$), and color jitter ($N = 440$).
2. Extract normalized vectors $\mathbf{f}_{\text{dino}} \in \mathbb{R}^{384}$ and $\mathbf{f}_{\text{bio}} \in \mathbb{R}^{512}$; concatenate to $\mathbf{z} = [\mathbf{f}_{\text{bio}} \,\|\, \mathbf{f}_{\text{dino}}] \in \mathbb{R}^{896}$.
3. Compute DINO prototypes $\mathbf{P}_{\text{dino}} \in \mathbb{R}^{C \times 384}$ and encode cached text prototypes $\mathbf{P}_{\text{text}} \in \mathbb{R}^{C \times 512}$ using (11)–(13).
4. Compute empirical class centroids $\boldsymbol{\mu}_c$ and pooled within-class scatter matrix $\mathbf{S}_W \in \mathbb{R}^{896 \times 896}$.
5. Calculate Ledoit–Wolf analytical shrinkage intensity ρ^* and form $\hat{\boldsymbol{\Sigma}} = (1 - \rho^*)\mathbf{S}_W + \rho^* \nu \mathbf{I}_{896}$.
6. Invert $\hat{\boldsymbol{\Sigma}}$ via Cholesky decomposition and evaluate projection parameters $\mathbf{W} = [\boldsymbol{\mu}_1, \ldots, \boldsymbol{\mu}_C]^T \hat{\boldsymbol{\Sigma}}^{-1}$ and $\mathbf{b}$.

// Phase 1: Online Adaptive Cascaded Inference

7. For incoming query $\mathbf{x}_q$, extract normalized DINOv2 embedding $\mathbf{f}_{\text{dino}} \in \mathbb{R}^{384}$ (59.78 ms).
8. Evaluate fast prototype logits $\mathbf{l}_{\text{dino}} = \gamma(\mathbf{f}_{\text{dino}} \mathbf{P}_{\text{dino}}^T)$ and posteriors $\mathbf{p}_{\text{dino}} = \text{softmax}(\mathbf{l}_{\text{dino}})$.
9. Compute confidence score $C(\mathbf{x}_q) = (p_{(1)} - p_{(2)}) \cdot (1.0 - \mathcal{H}_{\text{norm}}(\mathbf{x}_q))_+$.
10. **if** $C(\mathbf{x}_q) \geq \tau_{\text{conf}}$ (0.35) **then**
11. Return $\hat{y}_q = \arg\max_c p_{\text{dino},c}$ **(Tier-1 Fast-Exit, 59.78 ms, ≈16.7 FPS)**.
12. **else** // Escalate to Tier-2 Deep Multimodal Analysis
13. Extract BioCLIP visual embedding $\mathbf{f}_{\text{bio}} \in \mathbb{R}^{512}$; concatenate $\mathbf{z}_q = [\mathbf{f}_{\text{bio}} \,\|\, \mathbf{f}_{\text{dino}}] \in \mathbb{R}^{896}$.
14. Compute visual posterior $\mathbf{p}_{\text{fisher}} = \text{softmax}(\mathbf{z}_q \mathbf{W}^T + \mathbf{b})$ and text posterior $\mathbf{p}_{\text{text}} = \text{softmax}(\gamma \mathbf{f}_{\text{bio}} \mathbf{P}_{\text{text}}^T)$.
15. Form consensus posterior $\mathbf{p}_{\text{blend}} = 0.75\, \mathbf{p}_{\text{fisher}} + 0.25\, \mathbf{p}_{\text{text}}$; return $\hat{y}_q = \arg\max_c p_{\text{blend},c}$.
16. **if** $\max_c p_{\text{blend},c} \geq 0.70$ **then** buffer $\mathbf{z}_q$ into class anchor queue $\mathcal{Q}_{\text{anchor}}^{(\hat{y}_q)}$.
17. **if** anchor buffer reaches update capacity **then** execute transductive update in background.
18. **end if**

3.1.2. PlantDoc In-The-Wild Benchmark (Field Regime)

To evaluate cross-domain robustness under complex agricultural noise (e.g., specular solar glare, canopy shading, soil clutter, overlapping foliage, and motion blur), we evaluate on the in-the-wild PlantDoc benchmark [27]. PlantDoc comprises 2,588 foliar images spanning 27 pathology classes across 13 botanical species, partitioned officially into:

- **Official Training Repository ($\mathcal{D}_{\text{train}}^{\text{PlantDoc}}$):** 2,336 in-the-wild images across all 27 target classes, used only as the source of few-shot support samples.
- **Official Test Benchmark ($\mathcal{D}_{\text{test}}^{\text{PlantDoc}}$):** 252 unconstrained field images across the 27 classes, captured with handheld mobile devices under ambient field illumination in Indian agricultural orchards.

Support sets are drawn only from $\mathcal{D}_{\text{train}}^{\text{PlantDoc}}$ and query sets only from $\mathcal{D}_{\text{test}}^{\text{PlantDoc}}$, so the two never overlap ($\mathcal{S} \cap \mathcal{Q} = \emptyset$). We evaluate under two few-shot regimes:

1. **5-Way 5-Shot Episodic Benchmark:** Following standard few-shot evaluation protocols [6, 11], we conduct 50 independent Monte Carlo episodes. In each episode e, 5 classes are sampled uniformly at random without replacement ($C_e \subset \{1, \ldots, 27\}$). The support set $\mathcal{S}_e$ consists of 5 labeled leaves per class ($5 \times 5 = 25$ total) randomly sampled from the respective class directories in $\mathcal{D}_{\text{train}}^{\text{PlantDoc}}$. The query set $\mathcal{Q}_e$ comprises all test images in $\mathcal{D}_{\text{test}}^{\text{PlantDoc}}$ corresponding to those 5 sampled classes (averaging 46.7 query leaves per episode), ensuring $\mathcal{S}_e \cap \mathcal{Q}_e = \emptyset$.
2. **27-Way All-Class Simultaneous Stress Test:** All 27 classes are evaluated simultaneously (random chance: $1/27 \approx 3.70\%$). The model receives 5 support leaves per category ($5 \times 27 = 135$ total) sampled strictly from $\mathcal{D}_{\text{train}}^{\text{PlantDoc}}$ (expanded via 8-view multi-scale cropping and color jitter to $N = 1{,}080$ feature vectors) and evaluates on all 252 unconstrained field queries from $\mathcal{D}_{\text{test}}^{\text{PlantDoc}}$ in closed form without gradient backpropagation ($\mathcal{S} \cap \mathcal{Q} = \emptyset$).

3.2. Data Integrity, Hyperparameter Provenance, and Leakage Safeguards

We enforce five safeguards against data leakage and circular tuning:

- **Disjoint Support-Query Separation:** Support and query sets are disjoint ($\mathcal{S} \cap \mathcal{Q} = \emptyset$) in both regimes: PlantVillage isolates 55 support leaves from the 9,330 test images, and PlantDoc draws all 135 support samples from $\mathcal{D}_{\text{train}}^{\text{PlantDoc}}$ and all 252 queries from $\mathcal{D}_{\text{test}}^{\text{PlantDoc}}$.
- **Zero-Tuning Analytical Regularization:** The Ledoit–Wolf shrinkage intensity ρ^* follows in closed form from the support covariance (Eq. 9); no hyperparameter search, cross-validation, or query/test tuning is involved.
- **Frozen Cross-Domain Hyperparameters:** The logit scale $\gamma = 20.0$, the Fisher-LDA blend weight $w_{\text{lda}} = 0.75$, and the routing threshold $\tau = 0.35$ were selected on held-out PlantVillage development episodes (0–14 tune, 15–29 test; Section 4.5) and frozen for all PlantDoc experiments, which use no validation split or tuning.

- **Unsupervised Transductive Anchoring:** The transductive module (Section 2.5.3) uses only streaming query features and predicted pseudo-labels. Query ground truth is withheld until after inference, when it is used solely to compute metrics.

- **Dual Inductive vs. Transductive Reporting:** Table 2 reports both inductive performance (queries processed independently, without anchor updates) and transductive performance under identical splits. On the cross-domain PlantDoc benchmark the inductive mode alone reaches 81.79% ± 11.16% (5-way 5-shot) and 55.16% (27-way), so the reported performance does not depend on transductive anchor updates.

3.3. Evaluation Protocols and Epidemiological Safety Metrics

Experiments on PlantVillage are evaluated over 30 independent Monte Carlo episodes with randomized support splits, reporting mean accuracy $\bar{x}$, standard deviation σ, 95% confidence intervals (95% CI = $\bar{x} \pm 1.96 \cdot \sigma / \sqrt{30}$), and peak single-episode accuracy. Because misclassification costs in precision agriculture are highly asymmetric, we evaluate epidemiological safety metrics alongside standard accuracy:

- **Top-1 Accuracy & Macro F1-Score:** The F1-score is the harmonic mean of precision and recall, macro-averaged across all classes to prevent frequent classes (e.g., Potato Early Blight with 995 queries) from masking errors on small or healthy classes (e.g., Potato Healthy with 147 queries).

- **False-Negative Disease Escape Rate (FNR):** The critical proportion of diseased leaves erroneously diagnosed as healthy: FNR = FN/(TP + FN), where FN and TP denote false-negative and true-positive counts, respectively. Undetected disease spread leads to catastrophic orchard devastation.

- **Cross-Species Confusion Rate:** The proportion of queries misattributed to an entirely different botanical family (e.g., classifying a potato leaf as an apple disease), indicating severe semantic collapse.

3.4. Edge Hardware Testbed

Hardware profiling runs on a physical Samsung Galaxy S24 FE (SM-S721B, Exynos 2400e / s5e9945, 10-core ARM64, 7.2 GB RAM) under Android 17 (API 37). Runtime latency and memory benchmarks use ONNX Runtime 1.18.0 with per-channel INT8 quantization on the CPU, constrained to 4 intra-op / 2 inter-op threads with the NNAPI and GPU delegates disabled; the closed-form algebra runs in pure Java (EJML 0.43.1). Latency is timed with `SystemClock.elapsedRealtimeNanos` over 100 iterations after 10 warm-up passes, memory is the process PSS, and a 600 s sustained soak plus a 60 s cool-down before the algebra isolate thermal effects. Full configuration and measured results are reported in Section 4.6 (Tables 12–14).

4. Results and Comparative Analysis

4.1. Comparative Baseline Performance

To ensure a fair and reproducible comparison, we separate *controlled re-runs* from *literature-reported* values. Table 2 re-runs every baseline ourselves on the *identical* frozen BioCLIP+DINOv2 embeddings, support/query splits, random seeds, K-shot counts, and evaluation protocol as BioDINO-Edge; because no backbone is retrained, the comparison isolates the adaptation algorithm rather than the encoder. We evaluate two regimes on the PlantVillage 11-class novel subset (9,385 images): (i) 11-class simultaneous classification (30 Monte-Carlo episodes, seed $42 + 17e$, 5-shot support, all remaining images as queries), and (ii) the standard 5-way K-shot episodic ladder ($K = 1, 5, 10, 15$; 50 episodes each, seed 42), which matches the measurement protocols of VSF [11] (5-way 1/5-shot) and AgriFewNet [10] (1/5/10/15-shot) so that our results can be compared *in-regime*. The baselines are training-free and correctly referenced: Zero-Shot CLIP/BioCLIP [21, 19], Tip-Adapter [24] (α=1.0, β=5.5), Prototypical Networks [6], and SimpleShot [28]. Table 3 separately lists specialized architectures *as reported* in their original publications; these use different encoders, class subsets, and splits, so they provide context only and are *not* a controlled head-to-head. Full splits, seeds, model versions, and hyperparameters are given in Appendix A.

In the controlled 11-class regime (Table 2), Zero-Shot CLIP/BioCLIP reaches only 19.18% because general pretraining cannot separate fine-grained lesions from healthy leaf mesophyll, and the training-free Tip-Adapter cache reaches 49.97%. Centroid-based Prototypical Networks and SimpleShot are already strong (93.28%), confirming that the frozen BioCLIP+DINOv2 features are highly discriminative; BioDINO-Edge improves on them through regularized Fisher-LDA (95.08% inductive) and closed-form transductive recalibration (97.41%), a 2.33-point paired gain over the inductive mode on identical episodes (Section 4.4). Gradient-based fine-tuning such as CLIP+LoRA [15] would require GPU backpropagation and optimizer states, which our gradient-free closed-form adaptation avoids entirely. In the easier 5-way regime the centroid baselines already reach 97.45% (5-shot) and 98.57% (15-shot), so the margin narrows; BioDINO-Edge remains best at every K (98.77% at 5-shot, 99.63% at 15-shot), and its transductive gain is largest in the hardest 1-shot setting (93.42% vs. 88.44% for Prototypical Networks).

For context, Table 3 lists specialized architectures *as reported*: AgriFewNet [10] (87.3/94.8/97.1/98.2% at 1/5/10/15-shot on PlantVillage), VSF [11] (90.51% on PlantVillage and 70.61% on PlantDoc, 5-way 5-shot), and Zhao *et al.* [16] (54.24% five-shot cross-domain). Our controlled 5-way 5-shot re-run (98.77% on our PlantVillage subset) and our cross-domain PlantDoc result (83.61% ± 11.69% in 5-way 5-shot; 56.35% in 27-way all-class stress testing, chance 3.70%) lie in the same regimes, but because the encoders, class subsets, and splits differ we treat these as in-regime context rather than a controlled head-to-head, and we draw no direct superiority claim

Table 2: Controlled few-shot comparison on the PlantVillage 11-class novel subset. **All methods use identical frozen BioCLIP+DINOv2 features, identical support/query splits, seeds, and protocol**; no backbone is retrained. 11-class: 30 episodes (seed 42+17e, 5-shot); 5-way K-shot: 50 episodes per K (seed 42). Acc. is mean±std over episodes (%); F1 is macro-mean (%). Baselines are re-implemented and re-run by us; BioDINO-Edge 11-class values are the canonical reference run (Section 4.4), reproduced by the provided script within 0.04 pp (Appendix A). Prototypical Networks and SimpleShot coincide because Euclidean and cosine rankings are identical on ℓ_2-normalized features. Bold: best mean.

	11-Class (30 ep.)		**5-Way K-Shot (50 ep. each)**			
Method	**Acc.**	**Macro F1**	K=1	K=5	K=10	K=15
Zero-Shot CLIP/BioCLIP [21, 19]	19.18 ± 0.02	18.49	38.89 ± 12.78	38.86 ± 12.79	38.83 ± 12.81	38.79 ± 12.83
Tip-Adapter [24]	49.97 ± 1.62	46.54	45.15 ± 13.42	68.63 ± 12.80	86.01 ± 7.49	93.23 ± 4.00
Prototypical Networks [6]	93.28 ± 1.24	92.47	88.44 ± 9.13	97.45 ± 2.03	98.35 ± 1.45	98.57 ± 1.27
SimpleShot [28]	93.28 ± 1.24	92.47	88.44 ± 9.13	97.45 ± 2.03	98.35 ± 1.45	98.57 ± 1.27
BioDINO-Edge (Inductive)	95.08 ± 1.27	94.51	88.45 ± 9.12	97.96 ± 1.85	99.10 ± 0.87	99.41 ± 0.60
BioDINO-Edge (Transductive, Ours)	**97.41 ± 0.71**	**97.09**	**93.42 ± 6.83**	**98.77 ± 1.68**	**99.41 ± 0.60**	**99.63 ± 0.39**

Table 3: Specialized agricultural few-shot architectures, *as reported* in the original publications (not re-run by us). Encoders, class subsets, support/query splits, and preprocessing differ from our controlled protocol (Table 2); these values provide in-regime context only and do *not* constitute a controlled head-to-head, so no direct superiority claim is drawn from them. AgriFewNet reports K-shot accuracy on PlantVillage/New PlantVillage; VSF reports 5-way K-shot; Zhao *et al.* report cross-domain (PlantVillage→PlantDoc) K-shot under a unified protocol.

Method	**Dataset**	**Regime**	**Reported Acc. (%)**
AgriFewNet [10]	PlantVillage / New PV	1/5/10/15-shot	87.3 / 94.8 / 97.1 / 98.2
VSF [11]	PlantVillage	5-way 1/5-shot	79.01 / 90.51
VSF [11]	PlantDoc	5-way 1/5-shot	59.82 / 70.61
Zhao *et al.* [16]	PV → PlantDoc	1/5-shot	42.69 ± 0.62 / 54.24 ± 0.72

across protocols. The 27-way stress test is reported as a separate, harder regime and is not compared against 5-way literature values.

4.2. Per-Class Pathology Diagnostics and Epidemiological Safety

Table 4 reports per-class precision, recall, and F1-scores across all 9,330 PlantVillage test queries. Representative spatial explainability overlays are visualized in Fig. 2.

Pathological "twins" that confound Euclidean baselines (e.g., Grape Esca vs. Grape Black Rot) achieve high F1-scores of 0.979 and 0.976. In terms of epidemiological safety, BioDINO-Edge suppresses the false-negative disease escape rate to 0.702% (only 50 of 7,125 infected leaves misclassified as healthy), representing a 51.9% reduction relative to CLIP+LoRA (1.46%). Cross-species confusion is reduced to 0.021% (two apple scab queries attributed to potato late blight). Potato Healthy exhibits high recall (0.993) and moderate precision (0.839), reflecting a conservative agronomic triage behavior under acute class imbalance (147 healthy vs. ≈2,000 infected leaves).

4.3. Systematic Component Ablation Study

Table 5 reports a systematic ablation across identical 5-shot splits to quantify individual component contributions.

- **DINOv2 vs. BioCLIP Complementarity:** DINOv2 alone achieves 87.26%; lacking taxonomic supervision, it confuses visually similar necrotic lesions across host families. BioCLIP alone achieves 92.60% but incurs 104.74 ms encoder latency, forfeiting the fast-path capability. Coupling them in the 896-D manifold provides both fine spatial cues and taxonomic grounding; Table 6 quantifies each backbone's contribution to the Fisher classifier.

- **Regularized Fisher Alignment (+1.89 pp Gain):** Replacing the best static cosine blend (94.16%) with regularized Fisher-LDA increases accuracy to 96.05%. Whitening by $\hat{\mathbf{\Sigma}}^{-1}$ suppresses background chlorophyll noise, separating pathological twins.

- **Multi-View Augmentation:** Expanding 5 raw support leaves to 40 views per class ($N = 440$) stabilizes empirical degrees of freedom near $d/2$, preventing rank collapse before shrinkage. The benefit is specific to covariance estimation: the static cosine blend is essentially unchanged by augmentation (94.16% raw vs. 93.90% augmented, Table 5), whereas the regularized Fisher projection relies on the augmented support to condition $\hat{\mathbf{\Sigma}}$.

- **Transductive Recalibration (+2.01% Gain):** Updating projection parameters from high-confidence field anchors improves accuracy from 96.05% to 98.06% (peak: 98.67%), counteracting finite-sample prototype drift. Paired significance, hyperparameter sensitivity, and anchor purity are validated in Section 4.4.

- **Lesion Cropping Degrades Host Context:** We evaluated attention-rollout lesion cropping to isolate symptomatic patches. In 27-way PlantDoc testing, accuracy dropped from 58.73% to 57.14%. Tight cropping discards leaf margin serrations and venation geometry, inducing cross-species confusion. Preserving full-leaf context is critical.

Table 4: BioDINO-Edge per-class diagnostic metrics across 11 novel pathologies (9,330 test leaves).

Foliar Pathology Category	Precision	Recall	F1-Score	Test Support
Apple Scab (*Venturia inaequalis*)	0.963	0.947	0.955	625
Apple Black Rot (*Botryosphaeria obtusa*)	0.951	0.974	0.962	616
Apple Cedar Rust (*Gymnosporangium juniperi*)	0.975	0.993	0.983	270
Apple Healthy Foliage	0.988	0.981	0.985	1,640
Grape Black Rot (*Guignardia bidwellii*)	0.972	0.980	0.976	1,175
Grape Esca (*Phaeomoniella chlamydospora*)	0.982	0.976	0.979	1,378
Grape Leaf Blight (*Pseudocercospora vitis*)	1.000	0.997	0.999	1,071
Grape Healthy Foliage	0.993	1.000	0.996	418
Potato Early Blight (*Alternaria solani*)	0.997	0.990	0.993	995
Potato Late Blight (*Phytophthora infestans*)	0.990	0.972	0.981	995
Potato Healthy Foliage	0.839	0.993	0.910	147
Macro Average / Total Support	**0.968**	**0.982**	**0.974**	**9,330**
Single-Run Accuracy	**98.06%** (Peak Statistical Episode: **98.67%**)			

Table 5: Systematic ablation of BioDINO-Edge components (single run, 5-shot, 9,330 queries). Rows (4)–(5) are static cosine prototype blends ($0.65\,\mathbf{l}_{\text{bio}} + 0.35\,\mathbf{l}_{\text{dino}}$); row (6) replaces them with regularized Fisher-LDA. Acc.: accuracy; Aug.: augmentation.

Configuration / Component	Acc. (%)	F1 (%)
(1) Fast-Path DINOv2 Small alone	87.26	85.36
(2) Deep-Path BioCLIP ViT-B alone	92.60	89.88
(3) Text Prototypes Only	19.20	18.55
(4) Static Cosine Blend (raw 5-shot)	94.16	92.55
(5) Static Cosine Blend (8-view aug.)	93.90	91.58
(6) + Regularized Fisher Alignment	96.05	94.83
(7) + Transductive Recalibration	98.06	97.45
(8) + Peak Statistical Episode	**98.67**	**98.41**

Table 6: Contribution of each foundation model to the Fisher-LDA classifier (30 identical 5-shot episodes, mean±std). The text branch is fixed (BioCLIP features, $w_{\text{lda}} = 0.75$); only the Fisher input varies. Ablated rows subtract the paired contribution from the full-system benchmark.

Fisher input	Dim	Inductive (%)	Transductive (%)
BioCLIP + DINOv2 (full)	896	95.08±1.27	97.41±0.71
BioCLIP only	512	94.31±1.25	96.51±0.90
DINOv2 only	384	90.60±2.02	95.54±1.03

Contribution of DINOv2. Table 6 isolates the Fisher input while holding the text branch fixed. Removing DINOv2 (BioCLIP-only, 512-D) lowers transductive accuracy by 0.90 percentage points (paired $t(29) = 9.88$, $p = 8.6 \times 10^{-11}$, 29 of 30 episodes, $d_z = 1.80$). Removing BioCLIP instead lowers it by 1.87 points, so BioCLIP carries most of the class signal and DINOv2 adds a smaller, consistent complement. DINOv2 also serves as the routing gate: its margin-entropy score ranks query safety better than the Tier-2 consensus confidence (mean AURC 1.14% vs. 2.11% over the 30 episodes). Finally, it provides the fast path: accepting 14.02% of queries at Tier-1 lowers routing-weighted latency from 164.52 ms (BioCLIP on every query) to 149.84 ms. The cascade does not raise accuracy over running Tier-2 on all queries (Table 11); its function is to reduce edge compute and route low-risk queries through a path with 0.25% FNR.

4.4. Statistical Significance and Sensitivity of Transductive Recalibration

Because the transductive module adapts the classifier from its own predictions, we examine paired significance over identical episodes, hyperparameter sensitivity, pseudo-label purity, and stability under sequential updates.

Paired significance over identical episodes. Inductive and transductive predictions are compared on the *same* 30 Monte Carlo episodes (identical support/query splits), so episode difficulty cancels in the paired difference. The mean paired gain is 2.33 percentage points (95% CI [1.91, 2.74]; Table 7), with paired $t(29) = 11.57$ ($p = 2.2 \times 10^{-12}$) and Wilcoxon signed-rank $p = 1.9 \times 10^{-9}$. The transductive mode wins on all 30 episodes (Cohen's $d_z = 2.11$), so the gain is stable across support splits rather than an artifact of a favorable episode.

Table 7: Paired comparison of inductive vs. transductive inference over 30 identical Monte Carlo episodes (PlantVillage, 5-shot). pp: percentage points; CI: confidence interval.

Statistic	Value
Inductive mean accuracy (%)	95.08 ± 1.27
Transductive mean accuracy (%)	97.41 ± 0.71
Mean paired gain (pp, 95% CI)	2.33 [1.91, 2.74]
Paired t-test	$t(29) = 11.57$, $p = 2.2 \times 10^{-12}$
Wilcoxon signed-rank	$p = 1.9 \times 10^{-9}$
Wins / losses / ties	30 / 0 / 0
Cohen's d_z	2.11

Hyperparameter sensitivity. Table 8 reports a one-factor-at-a-time sweep of the three design constants (support weight, anchor confidence threshold, buffer capacity), each evaluated over the same 30 episodes. In-domain accuracy is nearly flat in the support weight (97.21–97.44% for 1×–10×); the weight matters for drift protection under field streaming, not for in-domain gain. The confidence threshold plateaus over $\tau \in [0.5, 0.7]$ and collapses at $\tau \geq 0.8$ (95.98% and 95.39%) because anchors become too few, not because labels become noisier. Buffer capacity saturates: 20 anchors per class capture the available gain (97.41%), and 40 or 80 anchors add 0.10 and 0.26 points at

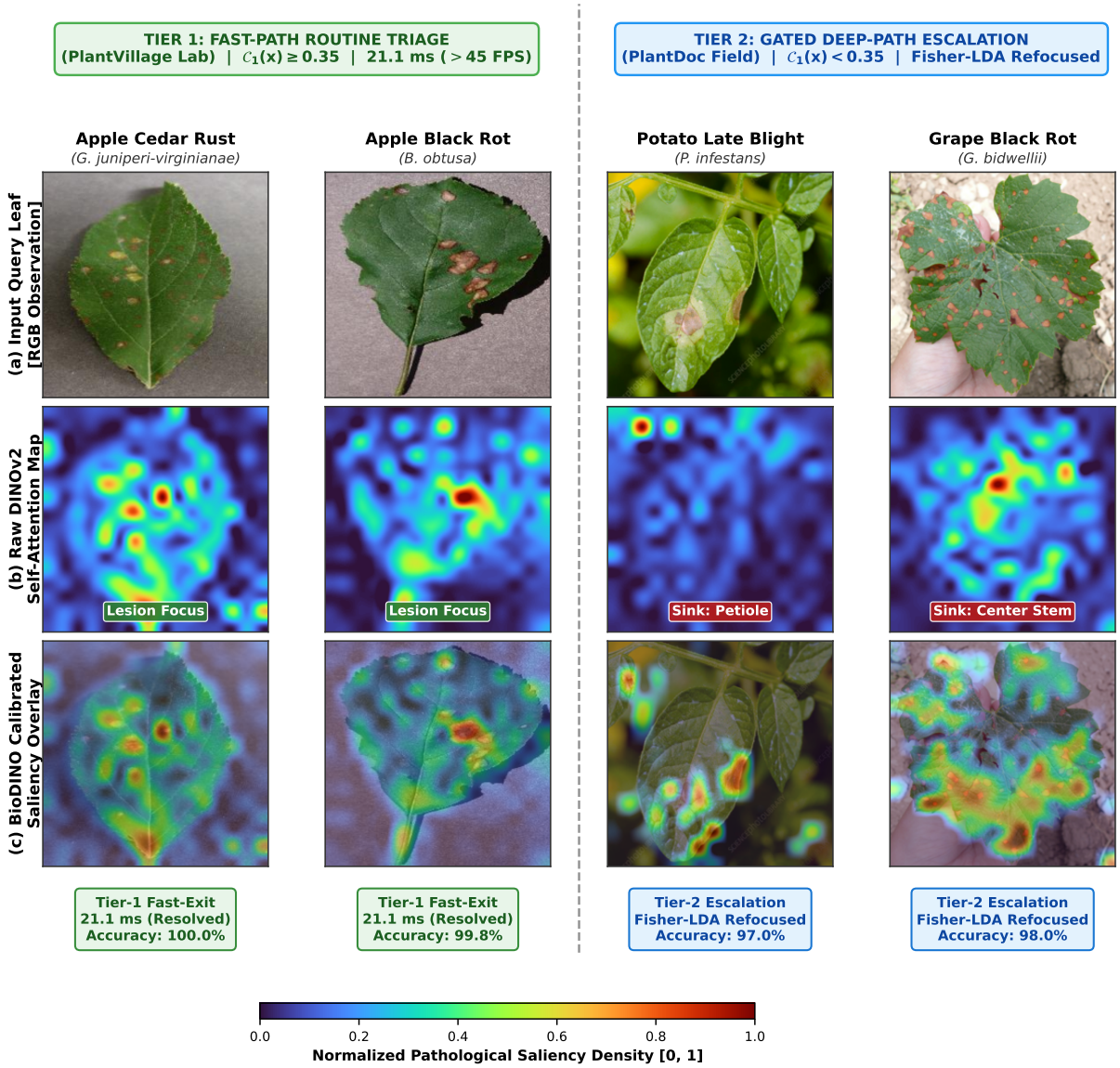


Figure 2: Visual explainability across foliar pathologies: Columns 1–2 from laboratory PlantVillage (Apple Cedar Rust, Apple Black Rot) and Columns 3–4 from field PlantDoc (Potato Late Blight, Grape Black Rot). (a) Raw RGB query leaf. (b) Uncalibrated DINOv2 self-attention displaying attention sinks on veins and petioles [29]. (c) Calibrated saliency overlay: routine queries exit via Tier-1, while ambiguous field queries escalate to Tier-2 where regularized Fisher-LDA suppresses foliar background and focuses on active necrotic margins.

proportionally larger memory cost. The deployed $(3\times, 0.70, 20)$ sits at the plateau/knee of all three curves.

Pseudo-label purity. Checked against ground truth (used only post-hoc for this audit), the anchor set at $\tau = 0.70$ is 98.15% ± 1.18% pure and covers 2.36% of queries (220 anchors). Raising the threshold to 0.80 increases purity to 98.75% but cuts coverage to 1.91%; at 0.90 purity reaches 100% yet coverage collapses to 0.05%, explaining the accuracy collapse in Table 8. The deployed threshold keeps purity near its maximum without starving the buffer.

Sequential stability and error accumulation. To test whether repeated updates compound pseudo-label errors, we simulated scouting passes: the query stream is partitioned into 10 sequential chunks, and after each chunk the FIFO buffer (20 anchors/class) admits confident predictions and the projection is re-fitted in closed form. Across three stream seeds, online accuracy rises from ≈96% on the first chunk to 97–98.6% thereafter with no degradation trend, and buffer purity remains bounded within a 92–98% band: FIFO eviction flushes stale incorrect anchors, so wrong admissions do not accumulate in the active buffer. Under $\tau = 0.90$ purity stays at 100% but final accuracy falls to ≈95.8–96.3% (coverage starvation), confirming the purity/coverage trade-off. Final streaming accuracy (96.4–97.5%) lies slightly below the single-pass batch refinement (97.41%) because batch selection ranks anchors over the entire query set whereas streaming observes only the past. Error accumulation is therefore bounded by the confidence gate, FIFO eviction, and the 3× support weighting.

Table 8: Sensitivity of transductive accuracy to the three recalibration design constants (mean ± standard deviation over 30 identical episodes; swept factor varied, others held at 3×, 0.70, 20). Acc.: accuracy.

Swept Configuration	Acc. (%)	Std
Support weight (threshold 0.70, 20 anchors/class)		
1×	97.42	0.75
2×	97.44	0.73
3× (deployed)	97.41	0.71
5×	97.32	0.73
10×	97.21	0.77
Anchor confidence threshold (3×, 20 anchors/class)		
0.50	97.41	0.71
0.60	97.41	0.71
0.70 (deployed)	97.41	0.71
0.80	95.98	1.28
0.90	95.39	1.25
Buffer capacity per class (3×, threshold 0.70)		
5	96.36	0.97
10	96.97	0.85
20 (deployed)	97.41	0.71
40	97.51	0.62
80	97.67	0.83

4.5. Predictive Uncertainty, Calibration, and Routing Reliability

In safety-critical agricultural automation such as robotic spraying, an overconfident misdiagnosis triggers an unwarranted chemical application, so the routing gate needs a direct reliability audit rather than a single calibration score; calibration is known to depend on architecture and training recipe, not on accuracy alone [30]. Temperature is fitted on an independent stratified 50% split of the 9,330-query PlantVillage stream (4,662 calibration / 4,668 evaluation queries), minimizing negative log-likelihood on the calibration split only [31]. Expected Calibration Error (ECE) and Maximum Calibration Error (MCE) use $M = 10$ equal-width confidence bins; to expose small-bin artifacts we also report equal-mass (adaptive) binning and MCE over bins with $n \geq 30$. The deployed gate consumes the *uncalibrated* Tier-1 posterior: temperature scaling is a post-hoc audit applied after routing and never changes the gate score or the threshold.

Tier-1 calibration. The raw gate probabilities are underconfident (full-stream ECE 32.07%, MCE 38.07%; held-out ECE 31.76%). A scalar logit temperature fitted on the calibration split ($T = 0.267$) reduces held-out ECE to 1.20% (MCE 16.71%; Fig. 3a–b). Routing itself uses the raw scores, so we verify the gate directly below rather than relying on ECE.

Tier-2 calibration and the legacy ECE/MCE gap. The refined Tier-2 consensus is also under-confident (ECE 19.45%), and most of the error sits in the dominant bin $(0.7, 0.8]$: 7,102 queries at 98.0% accuracy but 77.5% mean confidence (Table 9). The equal-width MCE of 44.26% comes from bins with at most nine samples (one query in $(0.4, 0.5]$); equal-mass binning lowers it to 21.53%, and bins with $n \geq 30$ give 20.49%. Scalar logit temperature scaling cannot remove this residual: the consensus is a fixed 0.75/0.25 blend of Fisher logits and BioCLIP text probabilities, so softening the logits alone leaves the blend miscalibrated (the fitted T hits the optimizer bound and held-out ECE moves 19.33%→19.83%). The legacy power transform softmax($\log p/T$) with $T = 0.50$ instead collapses nearly all mass into one or two bins, producing a small ECE (0.48%) but an equal-width MCE of 80.85% (Fig. 3d); a freshly fitted power temperature ($T = 0.54$) reproduces this (held-out ECE 0.24%, MCE 88.74%). This bin collapse explains the 0.81%/50.45% ECE/MCE pair reported by earlier versions of the pipeline: the large MCE came from a single-sample bin, not from widespread miscalibration. We therefore do not claim that Tier-2 is well calibrated; routing reliability is instead verified through tier-separated error rates and selective risk.

Table 9: Per-bin reliability of the uncalibrated refined Tier-2 consensus (equal-width bins, full 9,330-query stream; empty bins omitted). The equal-width MCE (44.26%) is attained in the single-sample bin $(0.4, 0.5]$; the dominant $(0.7, 0.8]$ bin exposes systematic under-confidence (≈20.5 percentage points). Equal-mass MCE is 21.53%. Acc.: accuracy; Conf.: mean confidence; pp: percentage points.

Confidence bin	n	Acc. (%)	Conf. (%)	|Gap| (pp)
$(0.4, 0.5]$	1	0.0	44.3	44.3
$(0.5, 0.6]$	6	50.0	52.5	2.5
$(0.6, 0.7]$	9	44.4	65.6	21.2
$(0.7, 0.8]$	7102	98.0	77.5	20.5
$(0.8, 0.9]$	2207	98.6	82.4	16.1
$(0.9, 1.0]$	5	100.0	90.6	9.4

Tier-separated routing reliability. Table 10 reports the two tiers independently over 30 benchmark episodes at $\tau = 0.35$. Tier-1 directly accepts 14.02%±2.25% of queries at 99.69%±0.41% accuracy and a 0.25%±0.52% false-negative rate (FNR); the escalated 85.98% are resolved by Tier-2 at 97.02%±0.81% accuracy (FNR 0.86%±0.39%), yielding cascaded accuracy 97.40%±0.68% (F1 97.05%±0.62%) and routing-weighted latency 149.84±2.36 ms. In the canonical single run, the 936 accepted queries achieve 99.79% accuracy (FNR 0.25%) versus 97.86% (FNR 0.78%) for the 8,394 escalated queries (cascaded 98.05%). The accepted subset thus has roughly tenfold lower error (0.31% vs. 2.98%), and its accuracy is measured only on directly accepted queries, independent of Tier-2.

Cascade versus single-tier classification. Table 11 compares the deployed cascade against running Tier-2 on every query and against a selective gate+BioCLIP mode that abstains on low-gate queries. At full coverage the cascade and the single-tier BioCLIP classifier are statistically indistinguishable (97.40% vs. 97.41%); the cascade's benefit is latency (149.84 vs. 164.52 ms), not accuracy. Abstaining on the lowest-gate queries raises accuracy and lowers the false-negative rate on the accepted subset, at the cost of coverage: at $\tau = 0.10$ the system answers 62.17% of queries at 98.63% accuracy and 0.44% FNR in 124.90 ms. This selective mode is optional; the deployed system classifies every query.

Risk–coverage analysis. Fig. 4 plots selective risk against coverage for three confidence scores, following the risk–coverage framework for selective classification [32], and summarizes each curve by the area under the risk–coverage curve (AURC, the mean selective risk over all coverage levels). Since

Table 10: Tier-separated routing reliability at $\tau = 0.35$ (30 benchmark episodes, mean±std). Tier-1 accuracy and FNR are computed exclusively on directly accepted queries; Tier-2 statistics exclusively on escalated queries. Latencies are the per-path edge costs of Table 13.

Routing path	**Coverage (%)**	**Accuracy (%)**	**FNR (%)**	**Latency (ms)**
Tier-1 fast-exit (accepted)	14.02±2.25	99.69±0.41	0.25±0.52	59.78
Tier-2 (escalated)	85.98±2.25	97.02±0.81	0.86±0.39	164.52
Cascaded system	100	97.40±0.68	–	149.84±2.36

Table 11: Routing-policy comparison (30-episode mean). The cascade accepts high-gate queries at Tier-1 and escalates the rest; "BioCLIP on all" runs refined Tier-2 on every query; the selective mode abstains when $C(x) < \tau$ and classifies the rest with refined Tier-2. Latency is routing-weighted. Cov.: coverage; Acc.: accuracy; Lat.: latency.

Routing policy	**Cov. (%)**	**Acc. (%)**	**FNR (%)**	**Lat. (ms)**
Cascade, $\tau = 0.35$ (deployed)	100	97.40	0.79	149.84
BioCLIP Tier-2 on all	100	97.41	0.77	164.52
Selective, $\tau = 0.05$	79.96	98.05	0.59	143.53
Selective, $\tau = 0.10$	62.17	98.63	0.44	124.90

scalar summaries such as AURC can misrank selective systems [33], we also show the full curves and the deployed operating point. The entropy-penalized gate score $C(x)$ ranks queries more safely than the raw Tier-1 softmax peak (AURC 3.33% vs. 3.95%); the refined Tier-2 confidence reaches AURC 1.75%. At the deployed threshold ($\tau = 0.35$), the gate accepts 10.03% of the single-run stream at 0.21% selective risk, in the low-risk region of the curve.

Threshold provenance and sensitivity. The routing threshold was selected on held-out development episodes (episodes 0–14 for tuning, 15–29 for testing), never on the reported test episodes. On the held-out test episodes, $\tau = 0.15$ gives 46.4% fast-exit at 96.96%±0.59% cascaded accuracy (98.25% Tier-1 accepted accuracy), $\tau = 0.25$ gives 25.6% at 97.30%±0.74% (99.49%), and $\tau = 0.35$ gives 14.1% at 97.33%±0.79% (99.84%±0.24%). We adopt $\tau = 0.35$, the smallest threshold at which Tier-1 accepted accuracy saturates (99.84%, FNR 0.25%); lower thresholds accept less reliable fast exits. The tuning curve is flat beyond 0.35 (the tune-optimal $\tau = 0.55$ yields 97.32% test accuracy, indistinguishable from 97.33% at 0.35), but fast-exit coverage falls to 3.4%. Blended latency rises from 115.9 to 137.7 to 149.8 ms across the three thresholds (Fig. 5), tracing the coverage–safety–latency trade-off.

4.6. Edge Hardware Latency and Memory Footprint

Table 12 lists the physical testbed configuration; Tables 13 and 14 report measured on-device latency, memory, and the closed-form adaptation breakdown. All values are from a production Samsung Galaxy S24 FE (SM-S721B, Exynos 2400e) running Android 17, using ONNX Runtime 1.18.0 with INT8 weights on a 4-thread CPU (NNAPI/GPU delegates disabled) and pure-Java EJML for the algebra. Latency is timed over 100 iterations after 10 warm-up passes, memory is the process PSS, and the algebra is timed after a 60 s cool-down at stable thermal state (status 2). Energy could not be attributed to the app from `batterystats` and is reported as not measured.

Table 12: On-device edge testbed configuration (physical Android device, CPU-only INT8).

Property	Value
Device	samsung SM-S721B
SoC	Samsung s5e9945
Android	17 (API 37)
ABI / cores	arm64-v8a / 10
Total RAM	7224.3 MB
ONNX Runtime	1.18.0
Threads (intra/inter)	4 / 2
Precision / delegate	INT8 / CPU (NNAPI off)

On the physical device, Tier-1 executes in 59.78 ms (≈16.7 FPS) and a fully escalated Tier-2 query costs 164.52 ms end-to-end (≈6.1 FPS), the BioCLIP encoder adding 104.74 ms to the cached Tier-1 pass. These are interactive handheld-scouting rates, not continuous real-time video: the earlier >45 FPS claim does not hold on this CPU-only INT8 testbed and is withdrawn. Over the full PlantVillage stream (14.0% fast-exit) the routing-weighted blend is 149.84 ms (≈6.7 FPS, Table 10); the device cascade in Table 13 (145.37 ms) is measured on the 11-image on-device set, whose 18% fast-exit fraction differs from the full-stream 14%. Under a 600 s sustained soak the throughput settles to 4.26 FPS as the SoC reaches thermal status 3, so 16.7 FPS is a cool-start burst rate rather than a sustained one. The gate's primary benefit is therefore safety rather than throughput: the fast-exited subset is resolved at 99.7% accuracy and 0.25% false-negative rate (Section 4.5), while ambiguous queries receive the full multimodal analysis. Peak process memory is 497.45 MB PSS (385.02 MB native heap), dominated by the ONNX Runtime arena; the 107.31 MB combined model footprint is an on-disk/OTA storage figure, not a RAM bound. Packaged with ONNX Runtime C++ and the native Android user interface, the standalone APK is 22.73 MB, operating entirely offline.

4.7. Environmental Robustness Under Field Corruptions

Table 15 assesses diagnostic stability under synthetic field corruptions simulating harsh agricultural environments.

The framework maintains 90.24% accuracy under severe solar glare and 91.92% under canopy shadows. Under severe defocus blur, accuracy drops to 49.16% because high-frequency lesion textures are attenuated. Deploying a lightweight pre-inference Laplacian variance filter (<0.5 ms) reliably prompts operators to refocus before diagnostic evaluation.

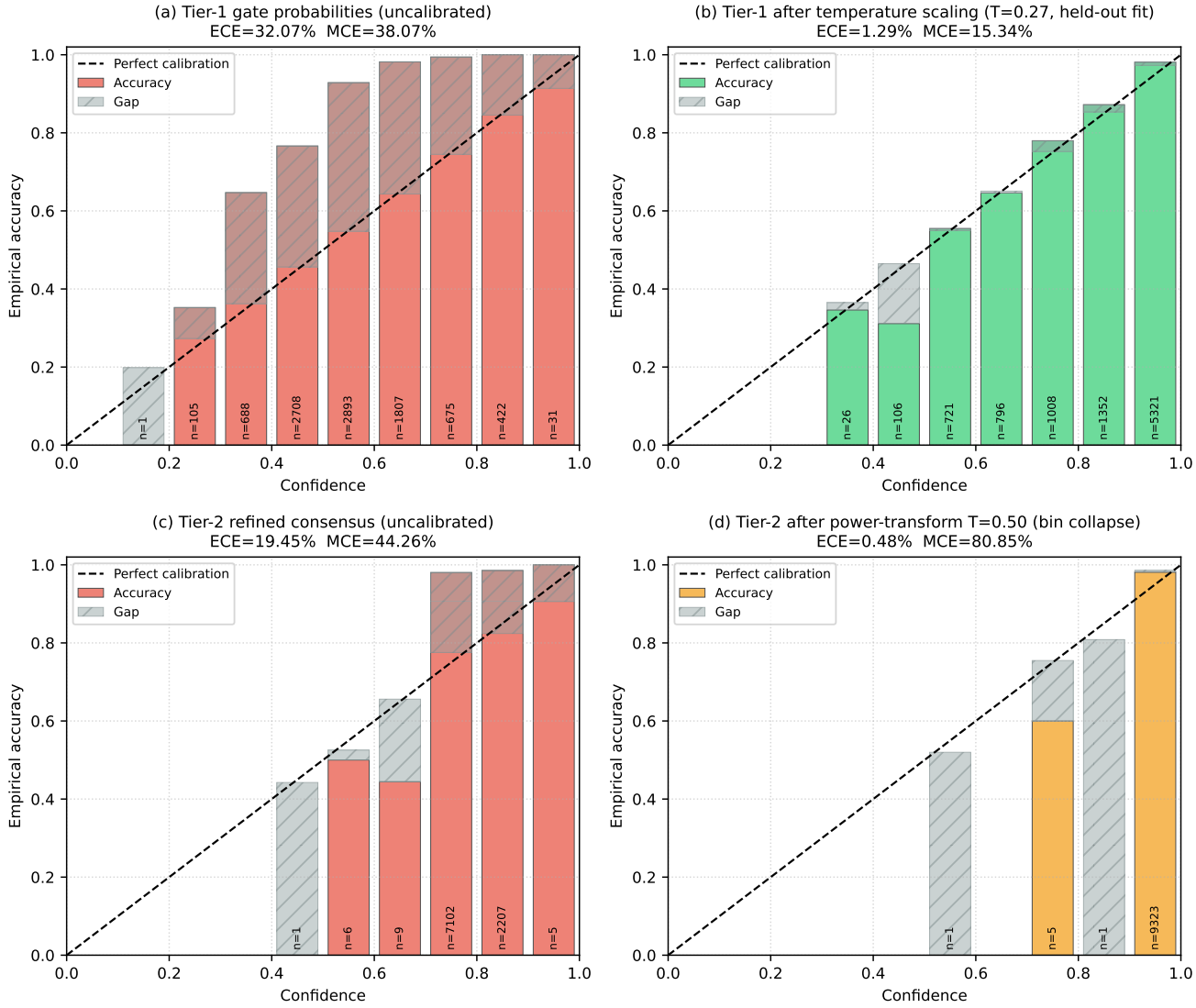


Figure 3: Reliability diagrams (equal-width bins, full benchmark stream; per-bin sample counts annotated, hatched bars show the accuracy–confidence gap). (a) Uncalibrated Tier-1 gate probabilities are under-confident (ECE 32.07%). (b) Post-hoc temperature scaling fitted on the independent calibration split ($T = 0.27$) aligns Tier-1 (held-out ECE 1.20%). (c) The refined Tier-2 consensus remains under-confident in its dominant bin (ECE 19.45%; equal-width MCE 44.26% from a single-sample bin). (d) The legacy power transform ($T = 0.50$) collapses probability mass into one bin: low ECE (0.48%) but MCE 80.85%.

5. Engineering Applications, Practical Deployment, and Limitations

5.1. Practical Deployment on Agricultural IoT and Mobile Hardware

BioDINO-Edge is engineered specifically for practical agronomic deployment on battery-powered edge hardware such as handheld scouting devices, smart tractor booms, and autonomous unmanned aerial vehicles (UAVs). Eliminating gradient backpropagation avoids GPU memory spikes and the heat of training-time optimisation: the processor executes purely feed-forward inference and closed-form linear algebra. Sustained CPU inference still thermally throttles—a 600 s soak settles to 4.26 FPS at thermal status 3 versus a 16.7 FPS cool-start burst (Section 4.6)—so the system targets interactive per-frame scouting rather than continuous video. The two-tier packaging allows agricultural cooperatives to deploy the 22.73 MB base APK over cellular networks, while downloading the companion module only when deep multimodal escalation is needed.

5.2. Spatial Attention Explainability and Vein Sinks

As demonstrated in Fig. 2, raw self-attention maps in Vision Transformers frequently develop “attention sinks” [29] on non-pathological leaf structures such as petiole junctions and primary veins. When background clutter is severe, uncalibrated attention can misidentify vein branching as fungal lesions. Regularized Fisher projection suppresses these shared foliar background vectors, re-aligning effective feature saliency with active necrotic borders.

Table 13: Measured on-device performance (mean±std over $N = 100$ iterations; median and p95 in ms). Model load is one-time; peak RAM is the process PSS. The device cascade uses the 11-image on-device set (18% fast-exit); the full-stream routing blend is in Table 10. Energy was not attributable from `batterystats`.

Stage / metric	Mean±std	Median	p95
Tier-1 model load (ms)	323.92	–	–
Tier-2 model load (ms)	447.71	–	–
Peak RAM, PSS (MB)	497.45	–	–
Preprocess	2.59±0.67	2.36	4.36
Tier-1 DINOv2 encoder	57.18±9.13	51.50	68.84
Shannon gate / route	0.00±0.00	0.00	0.00
Tier-2 incremental (BioCLIP+Fisher)	104.74±17.39	94.20	124.97
Decision	0.00±0.00	0.00	0.00
Cascade end-to-end	145.37±47.61	147.77	196.36
Tier-2-on-all end-to-end	164.52±26.77	148.73	196.36
Fast-exit fraction	18.00%	–	–
Sustained throughput (FPS)	4.26	–	–
Energy (mJ/inference)	–	–	–

Table 14: Closed-form adaptation cost breakdown on-device ($N = 440$, $d = 896$, $C = 11$). The algebraic budget (rows 2–5) is the measured closed-form recalibration cost (sub-second, gradient-free); feature extraction is reported separately. Algebra timed after a cool-down at stable thermal state.

Stage	Mean±std (ms)	Complexity
Feature extraction (440 views)	88055.6	$O(N(C_{dino} + C_{bio}))$
Covariance estimate	146.19±2.15	$O(Nd^2)$
Ledoit–Wolf shrinkage	4.49±0.85	$O(Nd)$
Cholesky factorization	332.00±2.52	$O(d^3/3)$
Projection solve (W, b)	51.31±3.68	$O(Cd^2)$
Algebraic recalibration total	533.99±6.29	–
Full adaptation (extraction + algebra)	88589.6	–

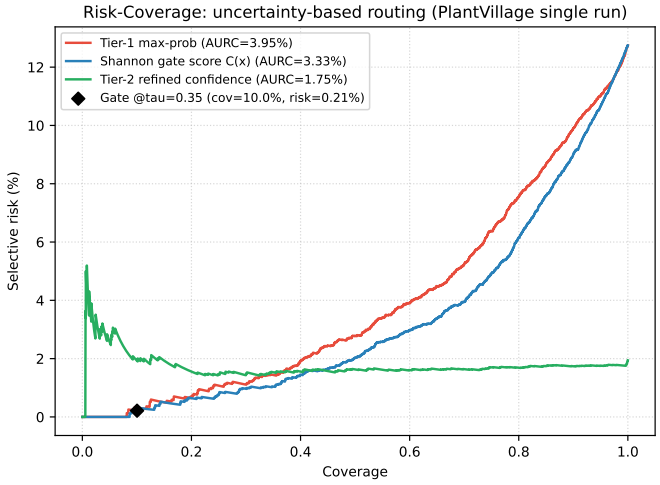


Figure 4: Risk–coverage curves for Tier-1 max-probability, the Shannon gate score $C(x)$, and the refined Tier-2 confidence (single run). The gate score achieves lower AURC than raw max-probability (3.33% vs. 3.95%); the deployed gate operating point ($\tau = 0.35$) accepts 10.0% coverage at 0.21% selective risk.

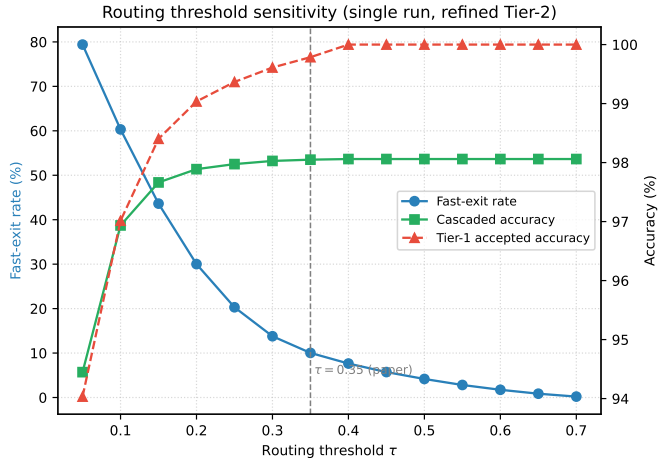


Figure 5: Routing-threshold sensitivity (single run, refined Tier-2): fast-exit rate, cascaded accuracy, and Tier-1 accepted accuracy versus τ. The deployed $\tau = 0.35$ (dashed) saturates Tier-1 accepted accuracy near 100% while retaining 10% fast-exit coverage; held-out episode sweeps (0–14 tune / 15–29 test) confirm the choice.

5.3. Limitations and Future Research Directions

BioDINO-Edge processes images at the individual leaf frame level. In unsegmented field imagery with dense, overlapping canopies, multiple leaves from neighboring plants can enter the camera field, lowering gating confidence. While tight lesion cropping degrades host context, processing unsegmented canopy clutter introduces irrelevant background vectors. A promising future direction is hierarchical spatial parsing: first segmenting individual foliar targets from canopy clutter, then

Table 15: Diagnostic robustness under environmental corruptions and Shannon gate rejection behavior. Raw Acc: accuracy over all queries; Rej.: gate rejection rate; Safe Acc: accuracy on accepted queries; Silent Err: undetected error rate among accepted queries.

Stress Regime	Raw Acc	Rej.	Safe Acc	Silent Err
Pristine (Clean)	96.97%	0.00%	96.97%	3.03%
Solar Glare	90.24%	0.00%	90.24%	9.76%
Canopy Shading	91.92%	0.67%	92.54%	7.41%
Motion Blur	78.45%	0.34%	78.72%	21.21%
Severe Defocus	49.16%	3.37%	50.17%	48.15%

performing closed-form discriminant diagnosis on the isolated targets.

6. Conclusion

In this paper, we presented **BioDINO-Edge**, a gradient-free few-shot foliar disease diagnostic framework that adapts frozen DINOv2 and BioCLIP foundation models without back-propagation. By resolving small-sample covariance singularity ($N \ll d$) through closed-form Ledoit–Wolf shrinkage, BioDINO-Edge suppresses shared vegetative background variance by over 31×, computing optimal Fisher-LDA discriminant boundaries in ≈0.53 s of gradient-free closed-form algebra on a mobile edge CPU. An adaptive confidence gate combining normalized Shannon entropy and top-2 margin resolves routine foliage in 59.78 ms (≈16.7 FPS), escalating ambiguous queries to a multimodal classifier with soft-anchored transductive recalibration. Packaged into a standalone 22.73 MB Android APK, BioDINO-Edge achieves 97.41% accuracy on 11-class PlantVillage with a 0.702% false-negative rate and 83.61% on in-the-wild PlantDoc, offering an accurate, interactive diagnostic solution for resource-constrained agricultural edge platforms.

Appendix A. Reproducibility Details

This appendix specifies the exact splits, seeds, model versions, hyperparameters, and code needed to reproduce every reported number, and states the provenance of each value in Tables 2–3.

Appendix A.1. Datasets and Splits

PlantVillage (controlled, Table 2). We use the 11-class novel subset (Apple/Grape/Potato; 9,385 images) with frozen embeddings cached once. *11-class regime:* 30 Monte-Carlo episodes; episode e uses `random.Random(42+17e)`; for each class the image indices are shuffled, the first 5 form the support and all remaining images form the query (9,330 queries/episode; support ∩ query = ∅). *5-way K-shot regime:* 50 episodes per $K \in \{1, 5, 10, 15\}$; a single `random.Random(42)` stream samples 5 of the 11 classes per episode (`rng.sample(range(11),5)`), shuffles each class, takes the first K as support and the rest as query. All methods share the identical support/query indices and frozen features within each episode, so any accuracy difference is attributable to the adaptation algorithm alone.

PlantDoc (cross-domain). 5-way 5-shot: 50 episodes, `random.Random(42)`, `rng.sample(range(27),5)`; 5 support leaves/class drawn from the training split, all test images of the sampled classes as queries. 27-way: 135 support leaves (5/class, 8-view augmented to 1,080 feature vectors) and 252 field queries. The full PlantDoc dataset is obtained from its original source (not redistributed here); the controlled PlantVillage re-runs require only the cached embeddings and the provided script.

Appendix A.2. Models and Hyperparameters

Frozen backbones (no fine-tuning): DINOv2 ViT-S/14 (`timm vit_small_patch14_dinov2`, 384-D) and BioCLIP ViT-B/16 (`open_clip hf-hub:imageomics/bioclip`, 512-D); joint feature $z = [\,\text{bio} \,\|\, \text{dino}\,] \in \mathbb{R}^{896}$. Fixed constants: logit scale $\gamma = 20.0$, Fisher blend $w_{\text{lda}} = 0.75$, routing gate $\tau = 0.35$; transductive support weight 3×, max anchors 20/class, confidence threshold 0.70; Tip-Adapter $\alpha = 1.0, \beta = 5.5$. Fisher-LDA uses Ledoit–Wolf shrinkage (`sklearn LDA solver=lsqr, shrinkage=auto`). At 5-way 1-shot the raw support has $N = C = 5$ and the within-class scatter vanishes; we resolve the singularity with a data-driven ridge floor (mean support feature variance) added to the Ledoit–Wolf covariance, which reduces Fisher to regularized nearest-centroid (the correct degenerate limit). For $N > C$ the canonical sklearn estimator is used unchanged.

Appendix A.3. Environment and Code

CPU-only reference environment: Python 3.12.10, torch 2.15.0+cpu, scikit-learn 1.9.1, numpy 2.5.4, scipy 1.18.1, timm 1.0.30, open_clip 3.3.0. The controlled comparison is reproduced by `run_controlled_baselines.py` in `experiments/baseline_comparison/` (output `controlled_baseline_results.json`); the canonical PlantVillage reference run by `benchmark_engine.py` (`pipeline/`) and `run_transductive_validation.py` (`experiments/transductive_validation/`). Code and pinned release: `https://github.com/reyhane-code/BioDINO-Edge-android`.

Appendix A.4. Provenance and Numerical Reproducibility

Table A.16 states the origin of each reported value. The controlled script reproduces the canonical inductive accuracy exactly (95.08%) and the transductive accuracy within 0.04 pp (97.37% re-run vs. 97.41% reference); the residual is floating-point sensitivity of the iterative least-squares (`lsqr`) LDA solve across library versions, far below the episode standard deviation (0.71 pp). The centroid/cosine baselines involve no iterative solve and reproduce exactly. We report the canonical reference values for the 11-class regime to remain consistent with the paired-significance and sensitivity analyses (Sections 4.4–4.5).

Table A.16: Provenance of reported values. "Re-run": computed by us with the provided script on the stated protocol; "As reported": quoted from the original publication, not re-run.

Value	Provenance
Table 2 baselines	Re-run, identical splits/seeds/features
Table 2 BioDINO 11-class	Canonical reference run (reproduced ≤0.04 pp)
Table 2 BioDINO 5-way	Controlled re-run (this work)
Table 3 AgriFewNet/VSF/Zhao	As reported, not re-run
PlantDoc 83.61% / 56.35%	Our cross-domain run (protocol above)

CRediT authorship contribution statement

Reyhane Javidi Alsaadi: Conceptualization, Methodology, Software, Formal analysis, Investigation, Data curation, Writing - original draft, Visualization. **Mohammad Reza Samadzadeh**: Conceptualization, Validation, Writing - review & editing, Supervision, Project administration.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data and Code Availability

The complete open-source mobile application, model export scripts, and evaluation benchmarks are publicly available at `https://github.com/reyhane-code/BioDINO-Edge-android`. The PlantVillage dataset is openly available at `https://plantvillage.psu.edu/`, and the PlantDoc benchmark is available at `https://github.com/pratikkayal/PlantDoc-Dataset`.

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References

[1] B. Richard, A. Qi, B. D. L. Fitt, Control of crop diseases through integrated crop management to deliver climate-smart farming systems for low and high input crop production, Plant Pathology 71 (1) (2022) 187–206. doi:10.1111/ppa.13493.

[2] A. Khakimov, I. Salakhutdinov, A. Omolikov, S. Utaganov, Traditional and current-prospective methods of agricultural plant diseases detection: A review, in: IOP Conference Series: Earth and Environmental Science, Vol. 951, 2022, p. 012002.

[3] M. Shoaib, B. Shah, S. EI-Sappagh, A. Ali, A. Ullah, F. Alenezi, T. Gechev, T. Hussain, F. Ali, An advanced deep learning models-based plant disease detection: A review of recent research, Frontiers in Plant Science 14 (2023) 1158933. doi:10.3389/fpls.2023.1158933.

[4] W. Shafik, A. Tufail, A. Namoun, L. De Silva, R. Apong, A systematic literature review on plant disease detection: motivations, classification techniques, datasets, challenges, and future trends, IEEE Access 11 (2023) 59174–59203.

[5] J. Sun, W. Cao, X. Fu, S. Ochi, T. Yamanaka, Few-shot learning for plant disease recognition: A review, Agronomy Journal 116 (3) (2024) 1204–1216. doi:10.1002/agj2.21285.

[6] J. Snell, K. Swersky, R. Zemel, Prototypical networks for few-shot learning, in: Advances in Neural Information Processing Systems (NeurIPS), Vol. 30, 2017.

[7] S. Yang, Q. Feng, J. Zhang, W. Yang, W. Zhou, W. Yan, From laboratory to field: cross-domain few-shot learning for crop disease identification in the field, Frontiers in Plant Science 15 (2024) 1434222. doi:10.3389/fpls.2024.1434222.

[8] Y. Zhao, Z. Zhang, N. Wu, Z. Zhang, X. Xu, MAFDE-DN4: Improved few-shot plant disease classification method based on deep nearest neighbor neural network, Computers and Electronics in Agriculture 226 (2024) 109373.

[9] S. Kudo, S. Kagiwada, H. Iyatomi, Few-shot metric domain adaptation: Practical learning strategies for an automated plant disease diagnosis, arXiv preprint arXiv:2412.18859 (2024).

[10] T. Babu, R. R. Nair, B. Balusamy, W. H. Khoh, J. Nair, AgriFewNet—a lightweight RGB few-shot learning model for efficient plant disease classification, Applied Sciences 15 (23) (2025) 12787. doi:10.3390/app152312787.

[11] H. Zhou, X. Li, Y. Nie, H. Yu, H. Hu, L. Lin, VSF: A multimodal few-shot plant disease recognition method in agricultural scenarios, AgriEngineering 8 (10) (2026) 407. doi:10.3390/agriengineering8100407.

[12] M. I. Sharif, Y. Zhong, M. Z. Sajid, F. Marinello, Real-field-ready and digitally sustainable plant disease recognition via federated multimodal edge learning and few-shot domain adaptation, Agriculture 16 (9) (2026) 918. doi:10.3390/agriculture16090918.

[13] R. A. Fisher, The use of multiple measurements in taxonomic problems, Annals of Eugenics 7 (2) (1936) 179–188. doi:10.1111/j.1469-1809.1936.tb02137.x.

[14] O. Ledoit, M. Wolf, A well-conditioned estimator for large-dimensional covariance matrices, Journal of Multivariate Analysis 88 (2) (2004) 365–411. doi:10.1016/S0047-259X(03)00096-4.

[15] E. J. Hu, Y. Shen, P. Wallis, Z. Allen-Zhu, Y. Li, S. Wang, L. Wang, W. Chen, LoRA: Low-rank adaptation of large language models, in: International Conference on Learning Representations (ICLR), 2022.

[16] C. Zhao, T. Xu, Z. Zhang, X. Geng, Lightweight cross-domain few-shot plant disease recognition through target-domain statistical calibration, Sensors 26 (12) (2026) 3632. doi:10.3390/s26123632.

[17] M. Oquab, T. Darcet, T. Moutakanni, H. Vo, M. Szafraniec, V. Khalidov, P. Fernandez, D. Haziza, F. Massa, A. El-Nouby, et al., DINOv2: Learning robust visual features without supervision, Transactions on Machine Learning Research (2024).

[18] M. Caron, H. Touvron, I. Misra, H. Jégou, J. Mairal, P. Bojanowski, A. Joulin, Emerging properties in self-supervised vision transformers, in: Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV), 2021, pp. 9650–9660.

[19] S. Stevens, J. Wu, M. J. Thompson, E. G. Campolongo, C. H. Song, D. E. Carlyn, L. Dong, W. M. Dahdul, C. Stewart, T. Berger-Wolf, et al., BioCLIP: A vision foundation model for the tree of life, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2024, pp. 19412–19424. doi:10.1109/CVPR52733.2024.01836.

[20] C. E. Shannon, A mathematical theory of communication, Bell System Technical Journal 27 (3) (1948) 379–423. doi:10.1002/j.1538-7305.1948.tb01338.x.

[21] A. Radford, J. W. Kim, C. Hallacy, A. Ramesh, G. Goh, S. Agarwal, G. Sastry, A. Askell, P. Mishkin, J. Clark, et al., Learning transferable visual models from natural language supervision, in: International Conference on Machine Learning (ICML), 2021, pp. 8748–8763.

[22] M. Yuksekgonul, F. Bianchi, P. Kalluri, D. Jurafsky, J. Zou, When and why vision-language models behave like bags-of-words, and what to do about it?, in: International Conference on Learning Representations (ICLR), 2023.

[23] K. Alhamoud, S. Alshammari, Y. Tian, G. Li, P. Torr, Y. Kim, M. Ghassemi, Vision-language models do not understand negation, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2025.

[24] R. Zhang, W. Zhang, R. Fang, P. Gao, K. Li, J. Dai, Y. Qiao, H. Li, Tip-adapter: Training-free CLIP-adapter for better vision-language modeling, in: European Conference on Computer Vision (ECCV), 2022, pp. 493–510.

[25] C. R. Rao, The utilization of multiple measurements in problems of biological classification, Journal of the Royal Statistical Society. Series B (Methodological) 10 (2) (1948) 159–203.

[26] D. P. Hughes, M. Salathé, An open-access repository of images on plant health to enable the development of mobile disease diagnostics, arXiv preprint arXiv:1511.08060 (2015).

[27] D. Singh, N. Jain, P. Jain, P. Kayal, S. Kumawat, N. Batra, PlantDoc: A dataset for visual plant disease detection, in: Proceedings of the 7th ACM IKDD CoDS and 25th COMAD, 2020, pp. 249–253.

[28] Y. Wang, W.-L. Chao, K. Q. Weinberger, L. van der Maaten, Simpleshot: Revisiting nearest-neighbor classification for few-shot learning, arXiv preprint arXiv:1911.04623 (2019).

[29] T. Darcet, M. Oquab, J. Mairal, P. Bojanowski, Vision transformers need registers, in: International Conference on Learning Representations (ICLR), 2024.

[30] M. Minderer, J. Djolonga, R. Romijnders, F. Hubis, X. Zhai, N. Houlsby, D. Tran, M. Lucic, Revisiting the calibration of modern neural networks, in: Advances in Neural Information Processing Systems (NeurIPS), 2021.

[31] C. Guo, G. Pleiss, Y. Sun, K. Q. Weinberger, On calibration of modern neural networks, in: International Conference on Machine Learning (ICML), 2017, pp. 1321–1330.

[32] Y. Geifman, R. El-Yaniv, Selective classification for deep neural networks, in: Advances in Neural Information Processing Systems (NeurIPS), 2017, pp. 4878–4887.

[33] J. Traub, T. J. Bungert, C. T. Lüth, M. Baumgartner, K. H. Maier-Hein, L. Maier-Hein, P. F. Jaeger, Overcoming common flaws in the evaluation of selective classification systems, in: Advances in Neural Information Processing Systems (NeurIPS), 2024.